Learning in Deep Artificial and Biological Neuronal Networks

Lecture #4: Learning Rules.



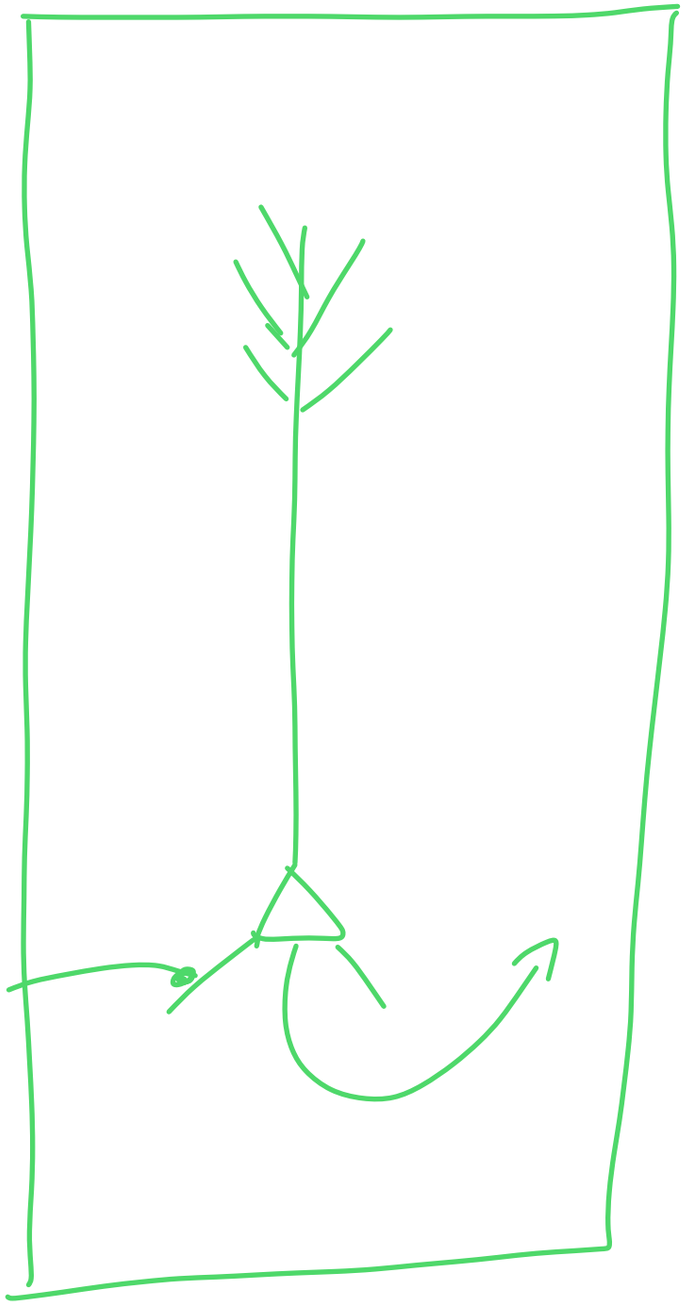
Lecture starts at 9:15am

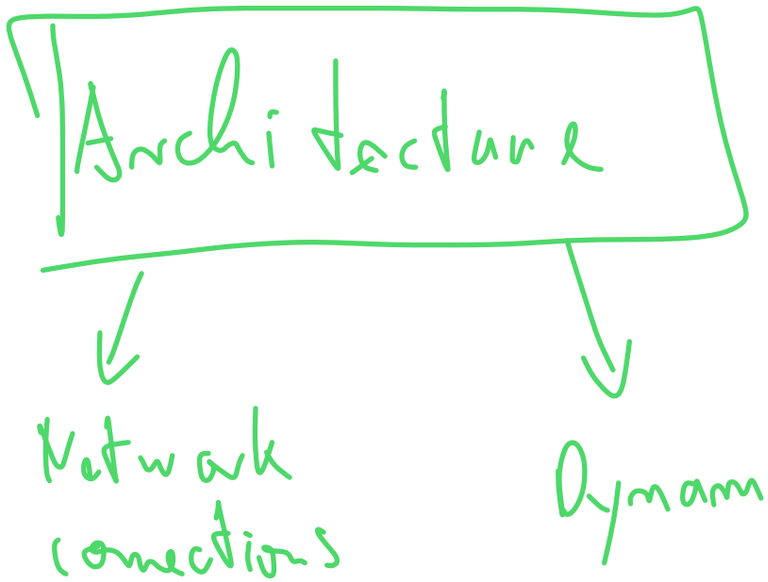
Benjamin F. Grewe
Institute of Neuroinformatics.
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Wednesday, Oct. 7th

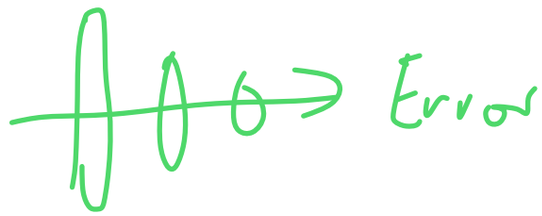
Neuronal (local) learning rules. - Why is this Topic important?

Levels ① ② ③





$$C = \frac{1}{2n} \| a - y \|^2$$



Learning Rules - Content Lecture #4.

1. Learning Rules:
   - 1.1) Why (local) neuronal learning rules important?
   - 1.2) The Perceptron Learning Rule
   - 1.3) ADALINE and the Delta rule
   - 1.4) Recap of Hebbian learning
   - 1.5) Oja's Rule
   - 1.6) The Covariance Rule
   - 1.7) Sanger's Rule
   - 1.8) The Calcium Rule
   - 1.9) The Bienenstock-Cooper-Monroe (BCM) rule.
   - 1.10) The Triplet Rule (Pfister&Gerstner).

Learning by error minimization.

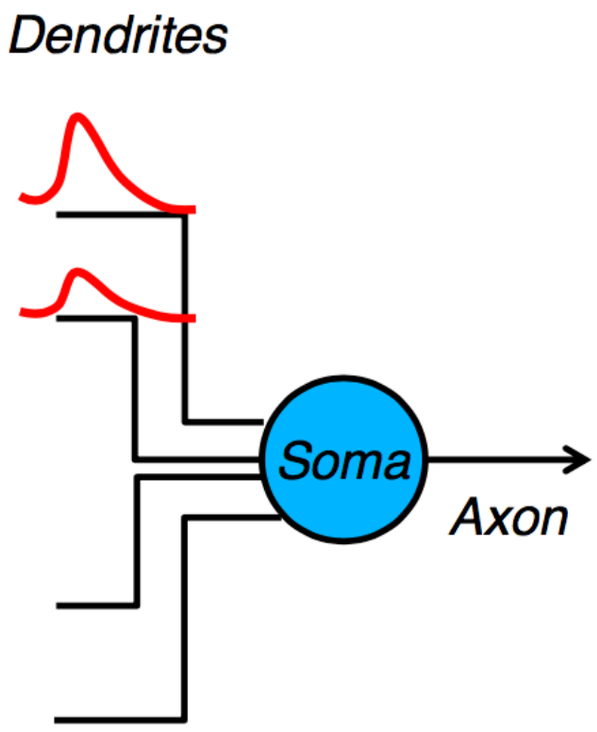
Bio Plausible

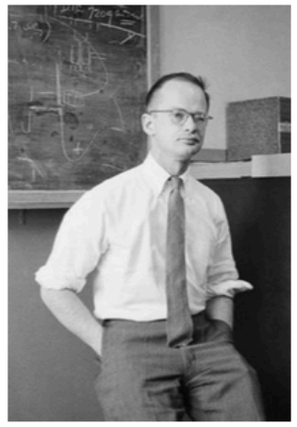
Variants of Bio plausible

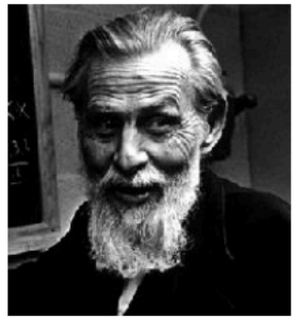
2. Extensions of hebbian learning rules (neo Hebbian)

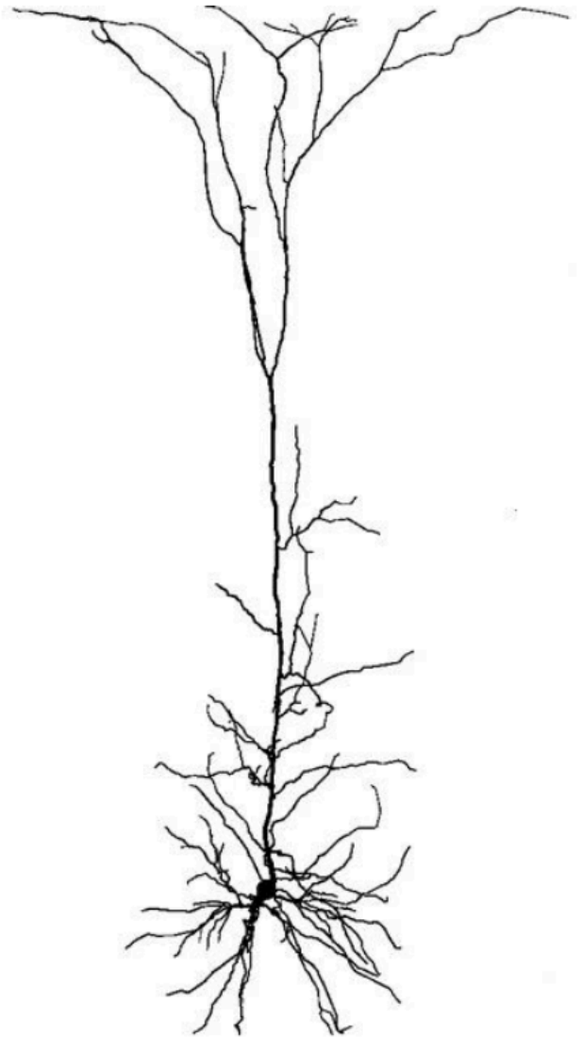
The Perceptron Idea has been Inspired by the Brain.

McCulloch and Pitts Neuron (MCP)







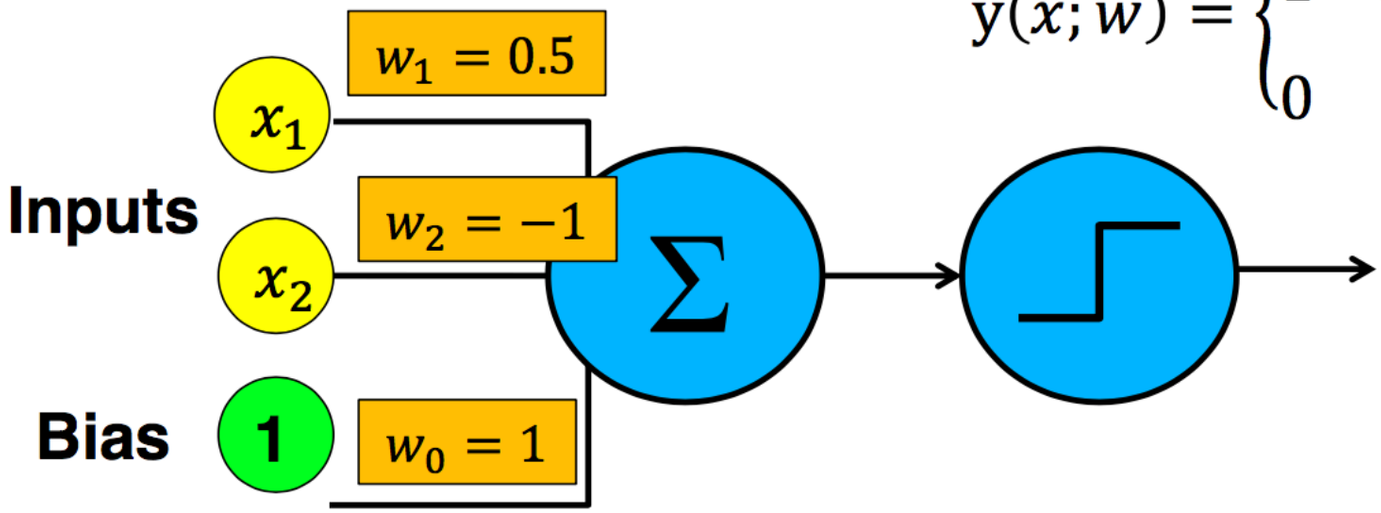


A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY*

■ WARREN S. MCCULLOCH AND WALTER PITTS
University of Illinois, College of Medicine,
Department of Psychiatry at the Illinois Neuropsychiatric Institute,
University of Chicago, Chicago, U.S.A.

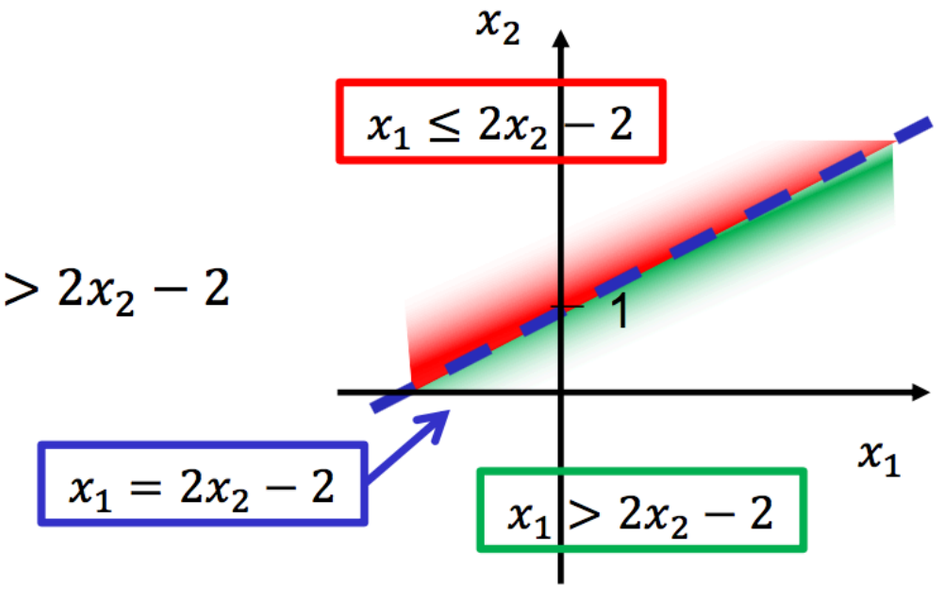
The Perceptron Implements a Linear Classifier.

$$y(\vec{x};\vec{w}) = \begin{cases} 1 & \text{if } \sum_{i=0}^{n} w_i x_i > 0 \\ 0 & \text{otherwise} \end{cases}$$

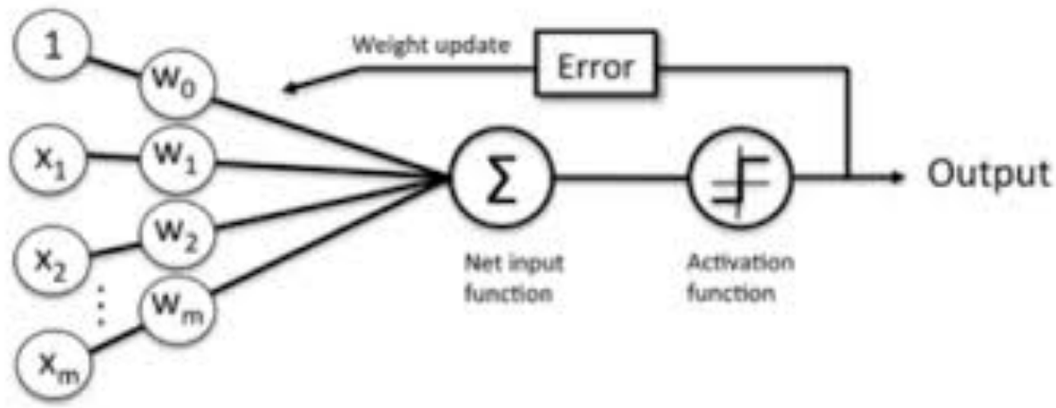


$$y(\vec{x};\vec{w}) = 1 \Leftrightarrow w_1 x_1 + w_2 x_2 + w_0$$
$$= 0.5\, x_1 - x_2 + 1 > 0 \Leftrightarrow x_1 > 2x_2 - 2$$

Decision Boundary:



Rosenblatt's Perceptron Learning Rule.



Algorithm: Perceptron Learning Algorithm

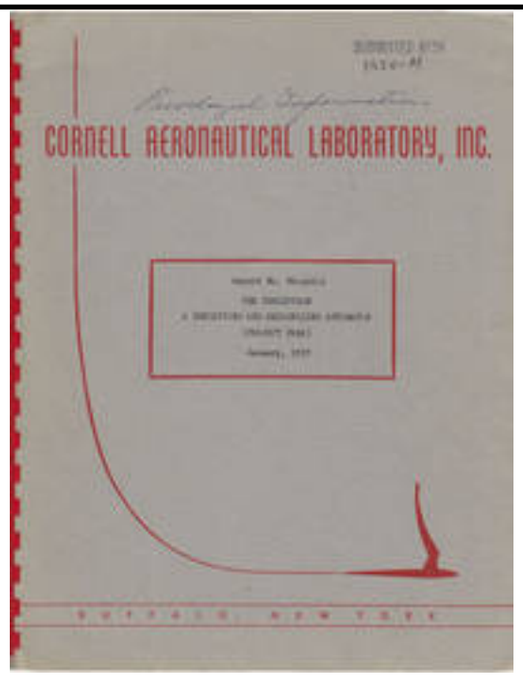
$P \leftarrow$ *inputs with label* 1;
$N \leftarrow$ *inputs with label* 0;
Initialize $\mathbf{w}$ randomly;
while !*convergence* **do**
  Pick random $\mathbf{x} \in P \cup N$;
  if $\mathbf{x} \in P$ *and* $\mathbf{w}.\mathbf{x} < 0$ **then**
    $\mathbf{w} = \mathbf{w} + \mathbf{x}$;
  end
  if $\mathbf{x} \in N$ *and* $\mathbf{w}.\mathbf{x} \geq 0$ **then**
    $\mathbf{w} = \mathbf{w} - \mathbf{x}$;
  end
end
//the algorithm converges when all the inputs are classified correctly

Target

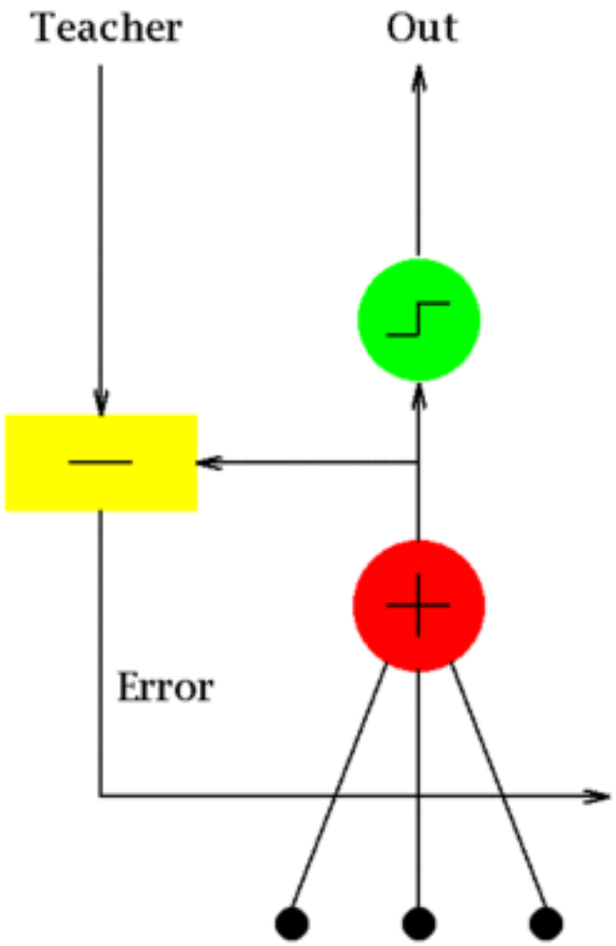
Perceptron Out

$$\Delta w_{ji} = \alpha (t_i - y_i) \cdot x$$

F. Rosenblatt. The perceptron, a perceiving and recognizing automaton Project Para. Cornell Aeronautical Laboratory, 1957.



ADALINE (ADAptive LInear NEuron)

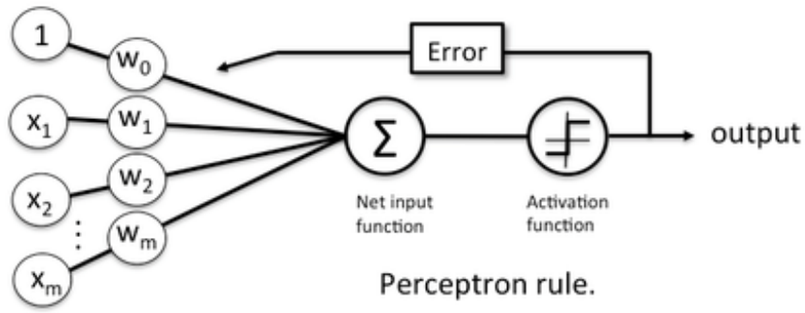


B. Widrow et al. Adaptive "Adaline" neuron using chemical "memistors". Number Technical Report 1553-2. Stanford Electron. Labs., Stanford, CA, October 1960.

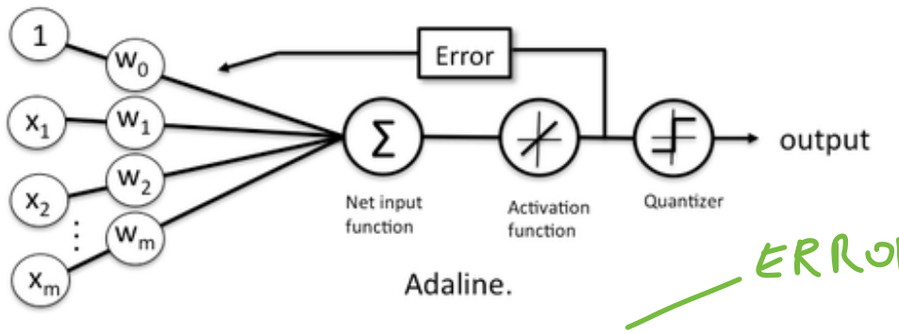
What do Adaline and the Perceptron have in common?

- They are both classifiers for binary classification.
- Both have a linear decision boundary.
- Both can learn iteratively, sample by sample (the Perceptron naturally, and Adaline via stochastic gradient descent).
- Both use a threshold function.

How do they differ?



Perceptron rule.

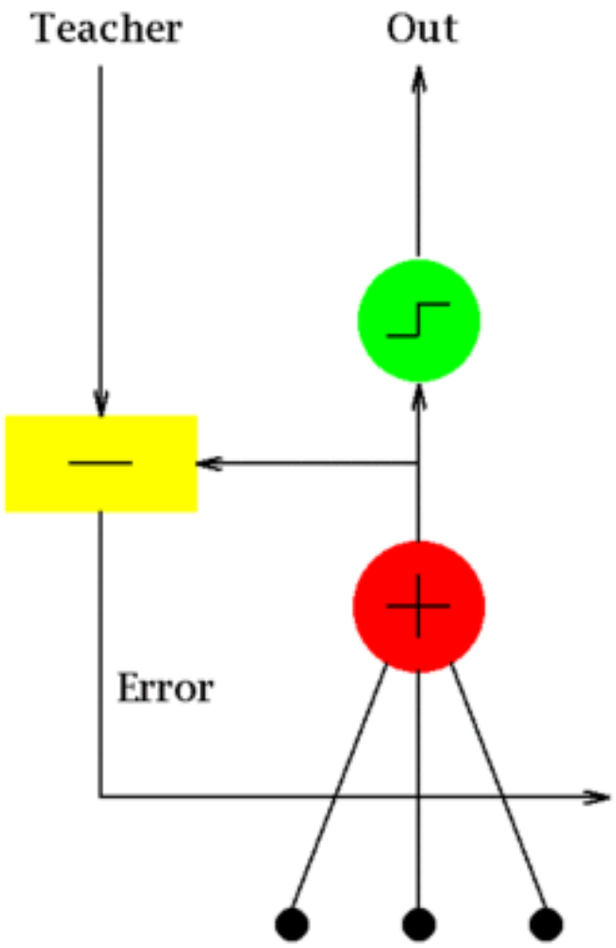


Adaline.

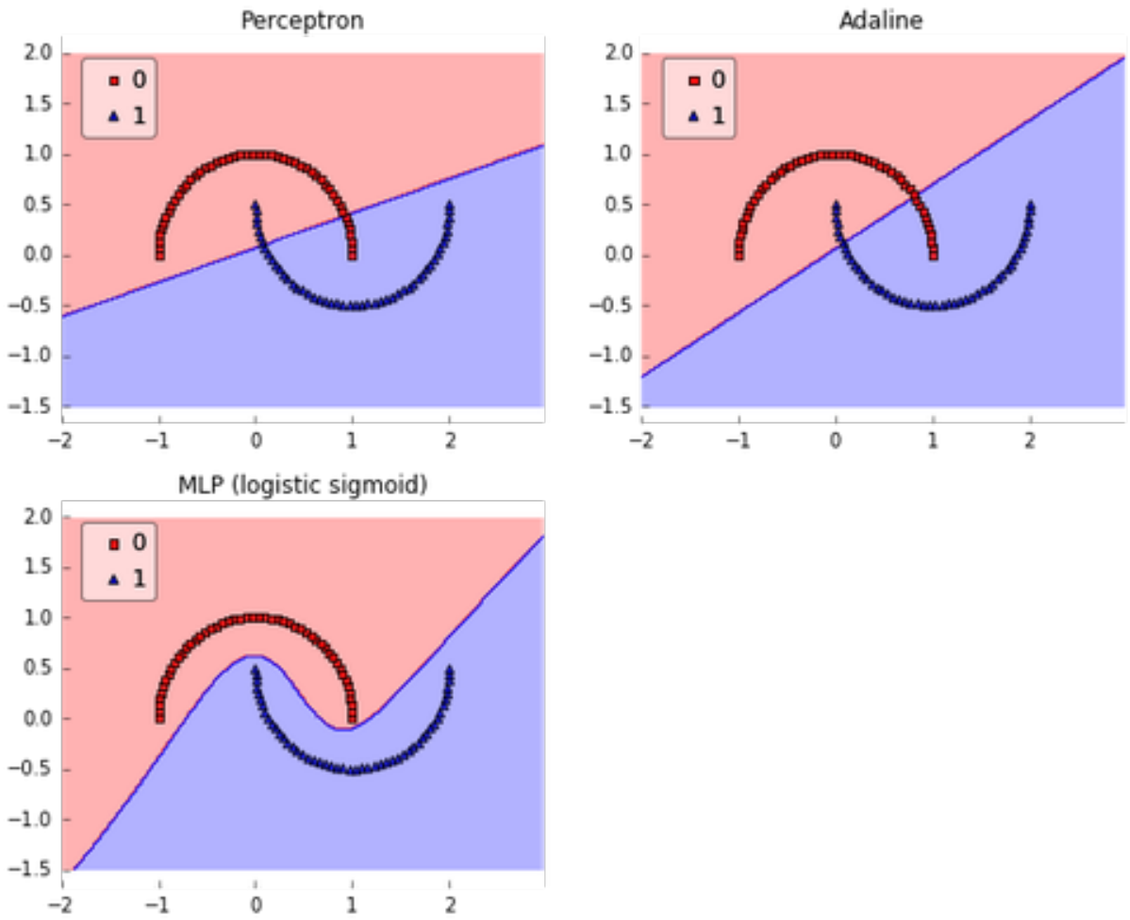
$$\Delta w_{ji} = \alpha (t_i - y_i) \cdot x_i$$

ERROR

ADALINE (ADAptive LInear NEuron)



What type of classifier can ADALINE implement?



The DELTA Rule

The Delta Rule uses the difference between target activation (i.e., target output values) and obtained activation to drive learning.

$$\Delta w_{ji} = \alpha \boxed{(t_i - y_i)} \cdot g'(h_j) x_i$$

The goal of the D-Rule is to minimize the output Error of the neuron.

α = learn rate
$g(x)$ = act. function
g' = deriv. of $g(x)$
t_i = target output
h_i = sum of inputs
y_i = output
x_j = input

Bernard Widrow und Marcian Edward Hoff: *Adaptive switching circuits*. IRE WESCON Convention Record, vol. 4, Los Angeles 1960, S. 96–104

The DELTA Rule

$$\frac{\partial E}{\partial w_{ji}} = \frac{\partial \left(\frac{1}{2}(t_i - y_i)^2\right)}{\partial w_{ji}} = \frac{\partial \left(\frac{1}{2}(t_i - y_i)^2\right)}{\partial y_i} \cdot \frac{\partial y_i}{\partial w_{ji}}$$

$$= -(t_i - y_i) \cdot \frac{\partial y_i}{\partial w_{ji}} = -(t_i - y_i) \cdot \frac{\partial y_i}{\partial h_i} \cdot \frac{\partial h_i}{\partial w_{ji}}$$

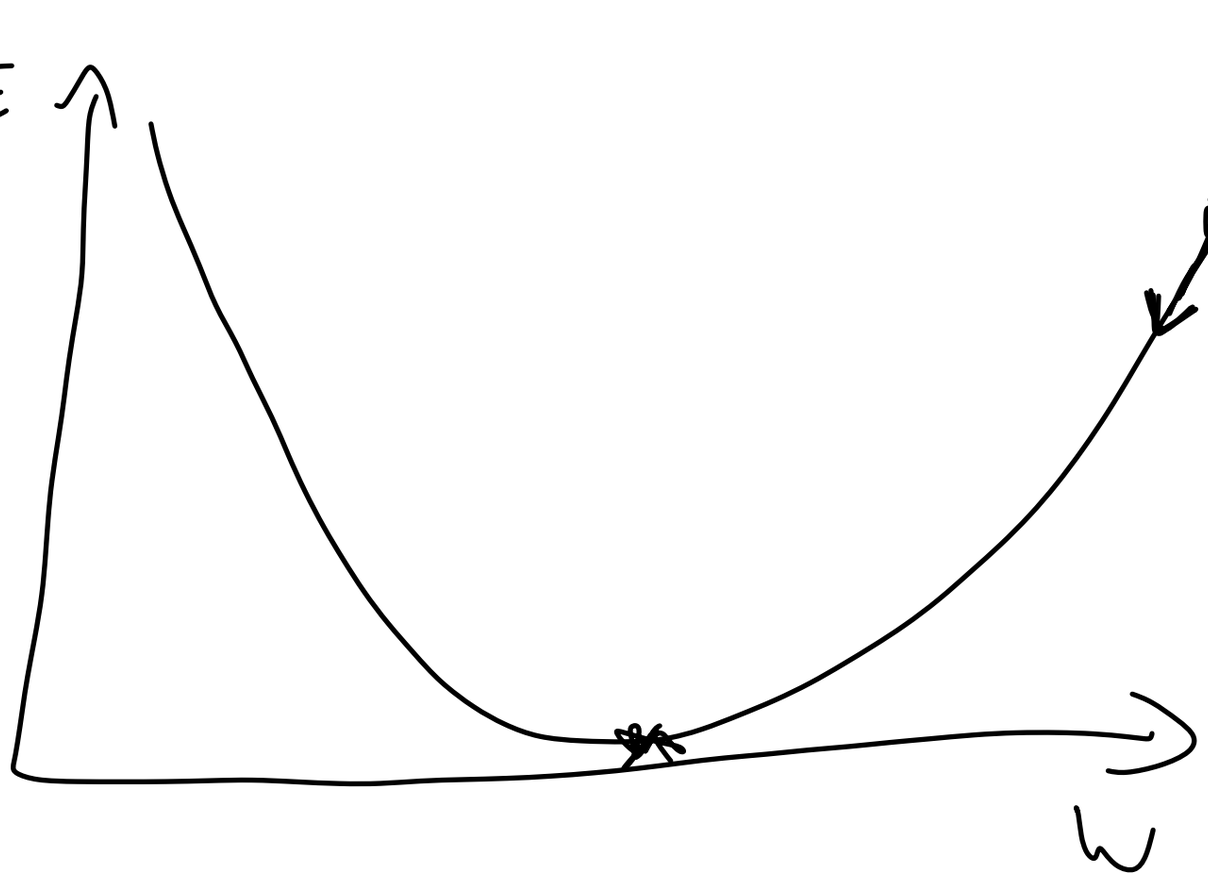
$g'(h_i)$

$$= -(t_i - y_i) \cdot g'(h_i) \cdot x_i$$

$$\frac{\partial \left(\sum_k x_k w_{jk}\right)}{\partial w_{ji}} = x_i$$

The DELTA Rule

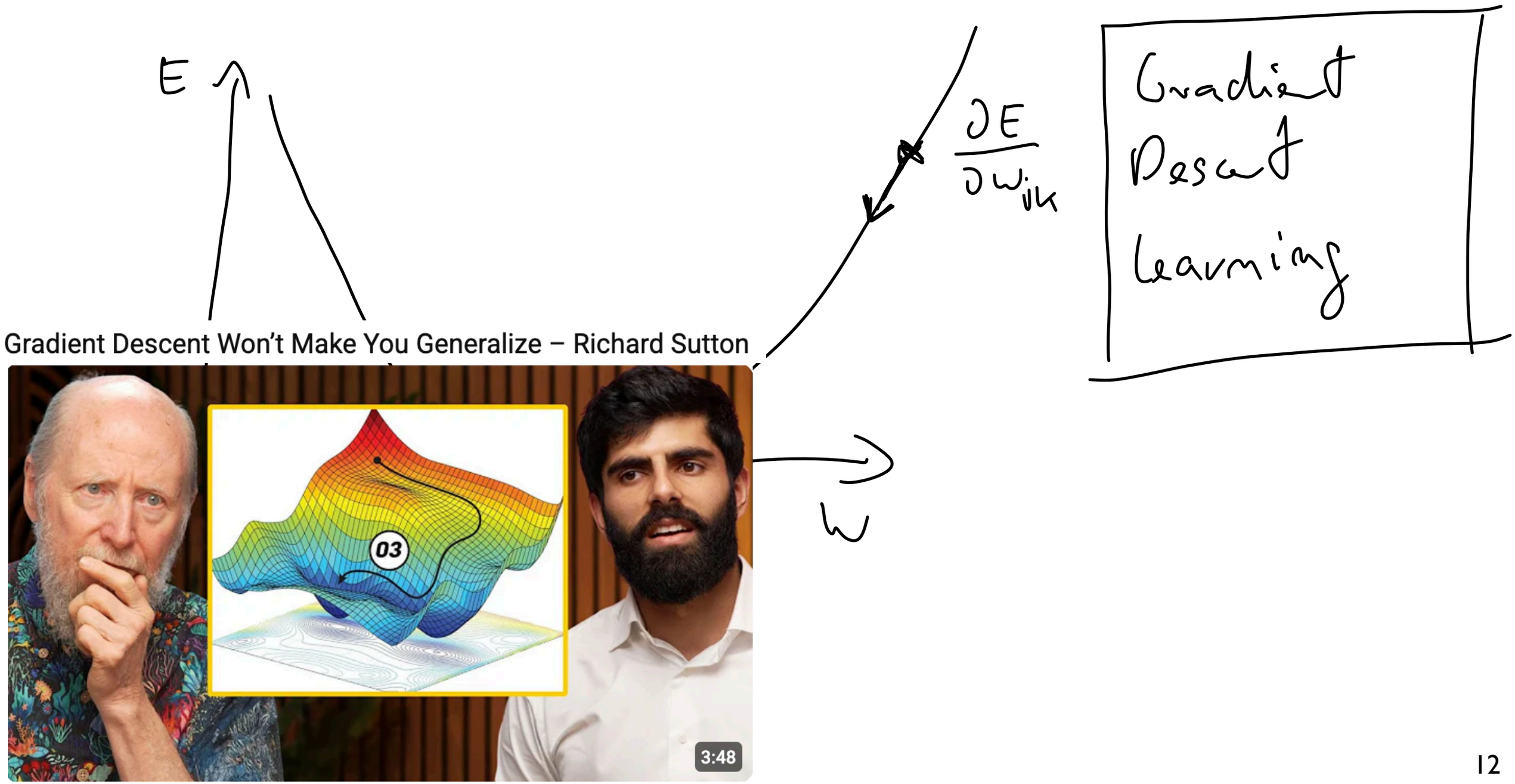
$$\Delta w_{ji} = \alpha (t_i - y_i)\, g'(h_i) \cdot x_i$$



Gradient Descent learning

The DELTA Rule

$$\Delta w_{ji} = \alpha (t_i - y_i)\, g'(h_i) \cdot x_i$$



DELTA Rule vs. Perceptron Learning Rule

We have seen that the Delta Rule and the Perceptron Learning Rule for training Single Layer Perceptrons have a similar weight update equation:

$$\Delta w_{ji} = \alpha (t_i - y_i)\, g'(h_i)\, x_i$$

Specfic reward expectations formed by **Rescorla-Wagner rule**

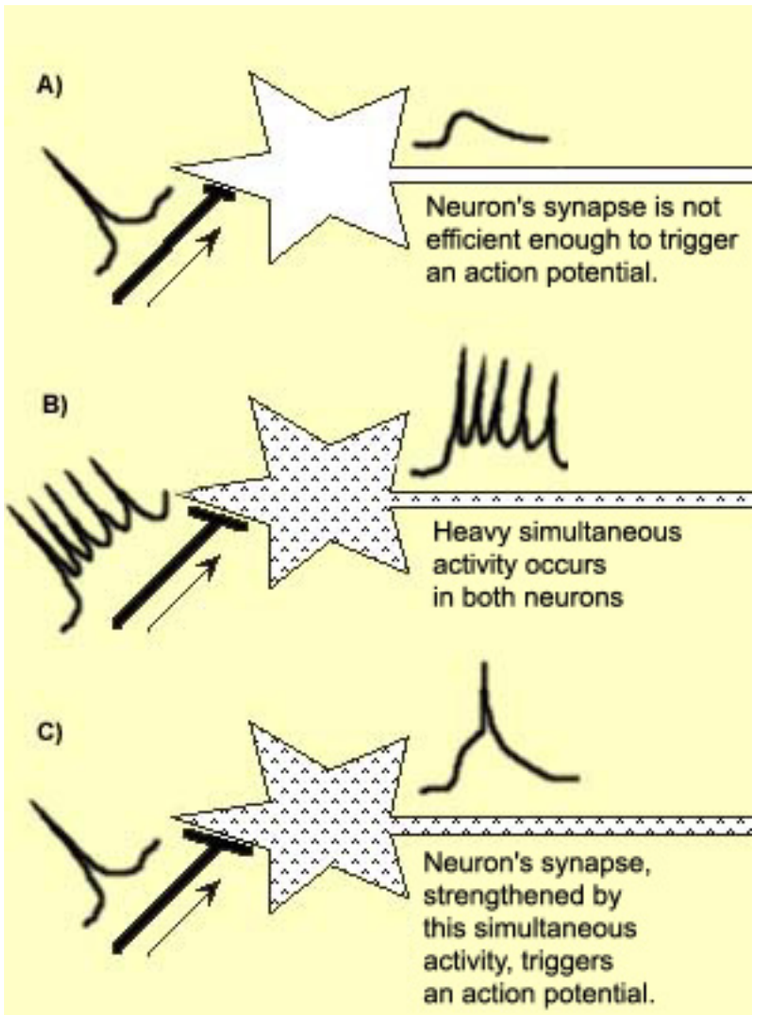
$$\boldsymbol{w} \rightarrow \boldsymbol{w} + \epsilon\, \delta\, \boldsymbol{u}, \qquad \delta = r - v$$

where ϵ is **learning rate** and δ is **prediction error**. For example, $\boldsymbol{u} = [0, 0, 0, 1]$:

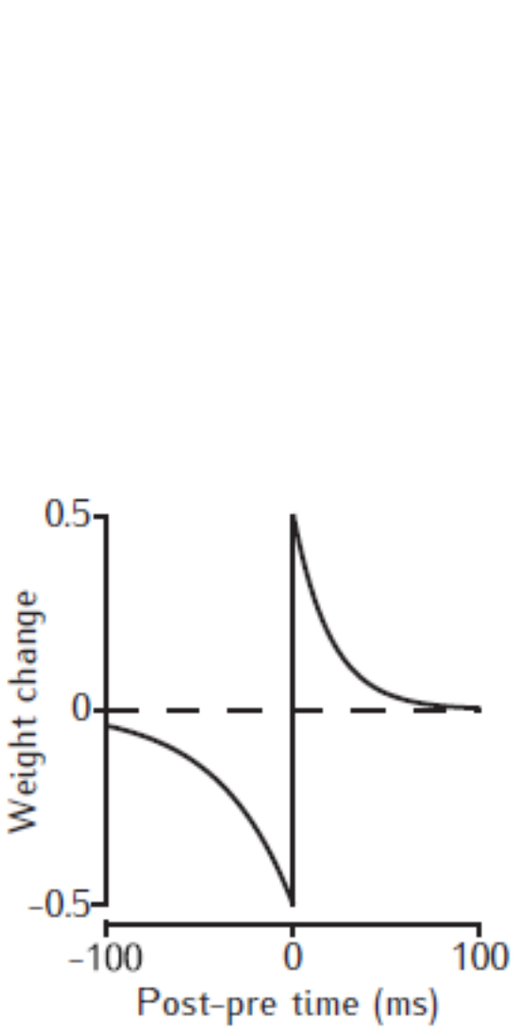
$$\begin{pmatrix} w_1 \\ \vdots \\ w_4 \end{pmatrix} \rightarrow \begin{pmatrix} w_1 \\ \vdots \\ w_4 \end{pmatrix} + \epsilon\, \delta \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} w_1 \\ \vdots \\ w_4 + \epsilon\, \delta \end{pmatrix}$$

Hebbian Learning and STDP

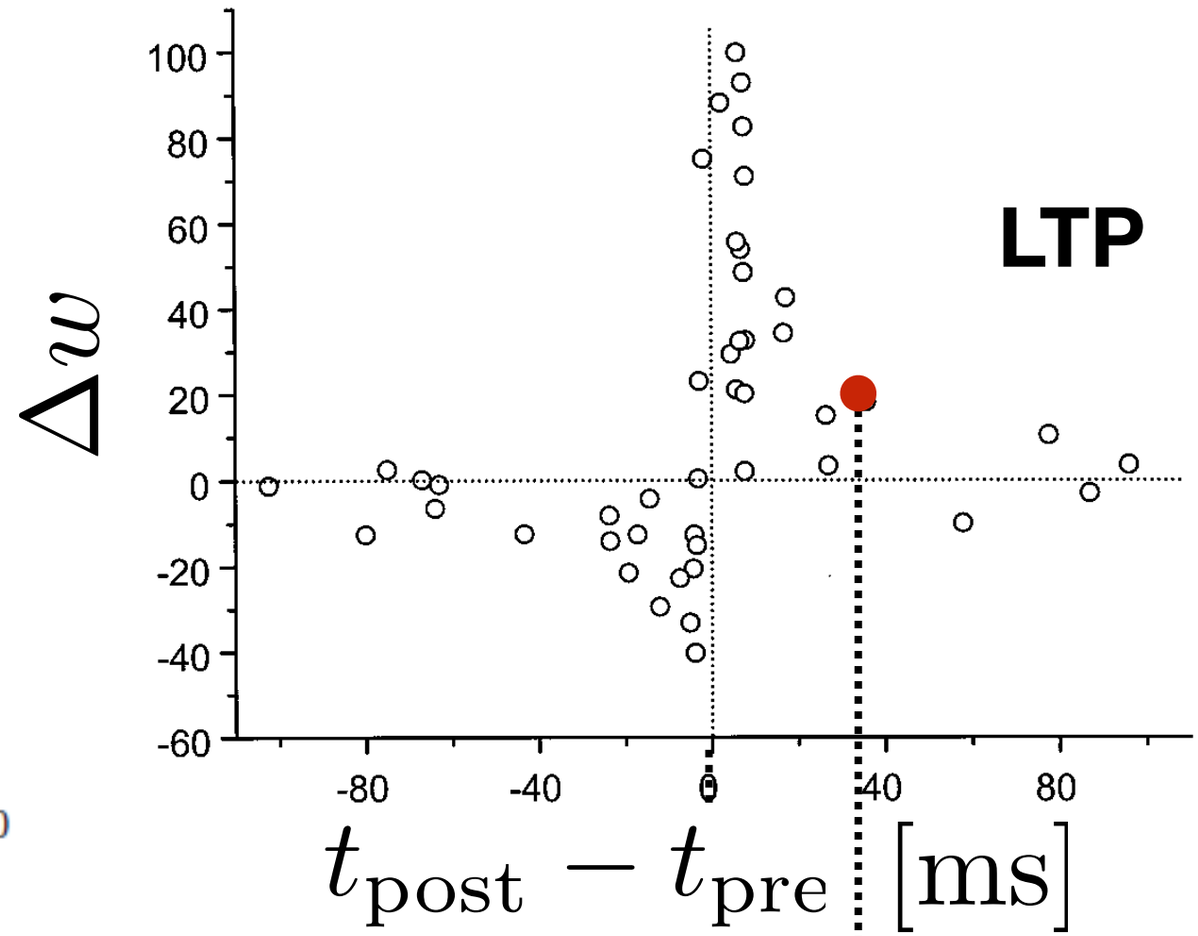
"When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place on one or both cells such that A's efficiency as one of the cells firing B, is increased."

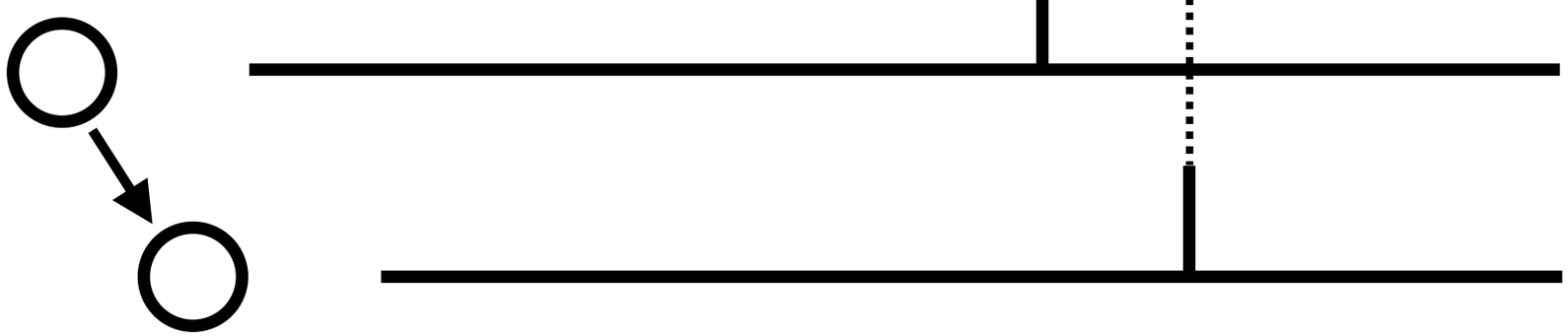


Examples of HL - Spike-timing Dependent Plasticity.



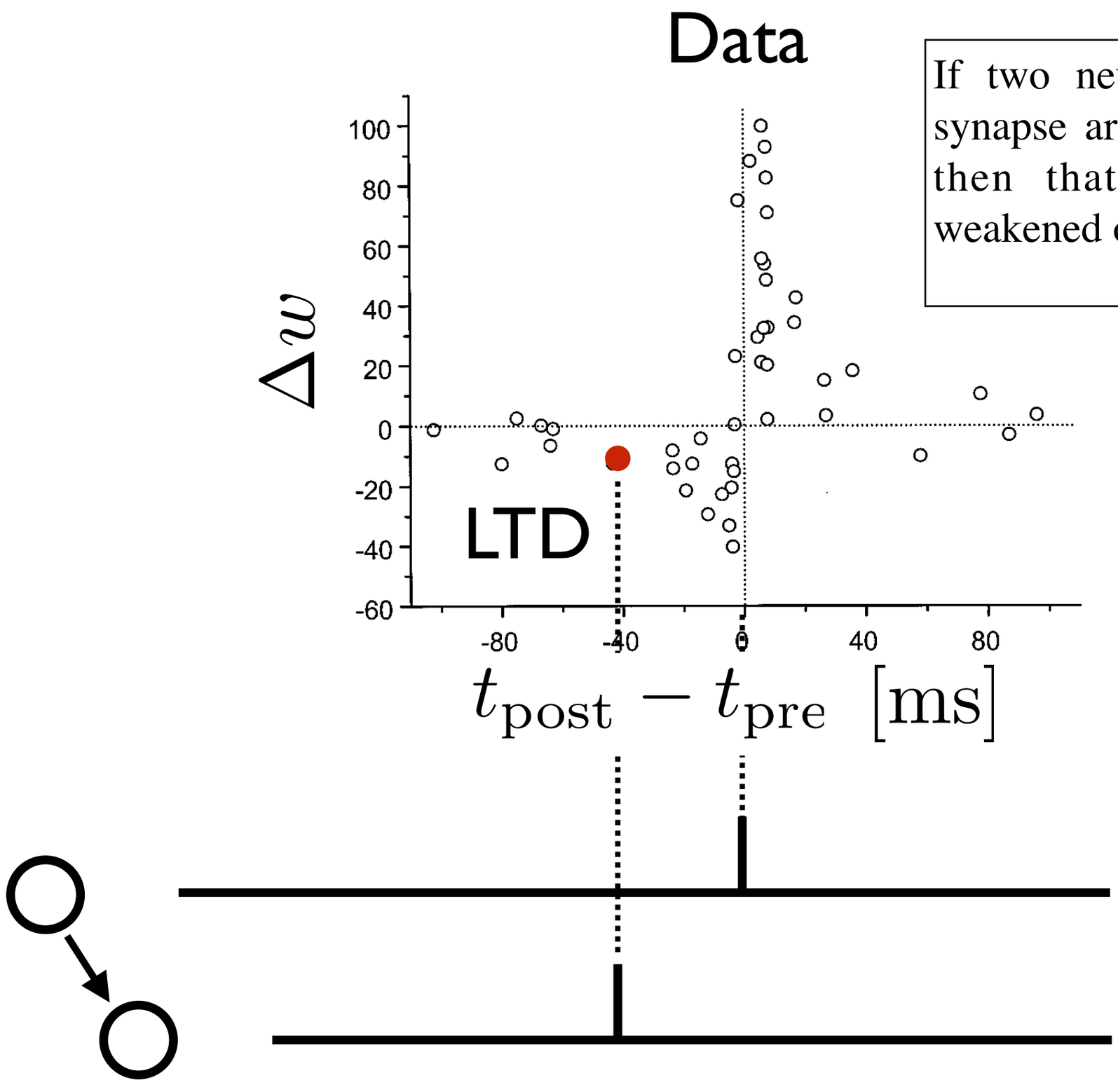
Data





Bi and Poo, 1998

Examples of HL - Spike-timing Dependent Plasticity.



If two neurons on either side of a synapse are activated asynchronously, then that synapse is selectively weakened or eliminated.

Bi and Poo, 1998

Hebbian Learning Rules

Hebbian Learning Rule:

$$\Delta w_{ij} = x_i \times y_j$$

x_i: input

y_j: output

x_i: presynaptic input to neuron

y_j: neuron output in form of a spike

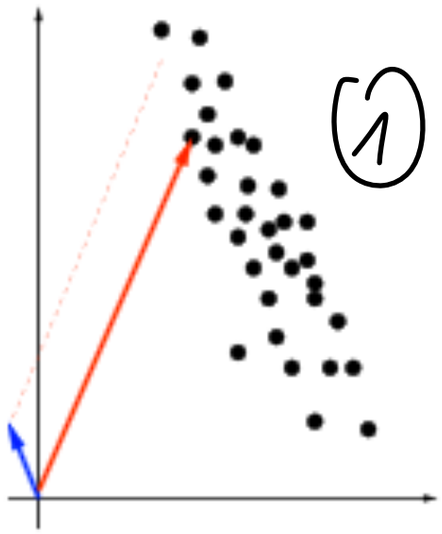
$$\Delta \bar{w} = \alpha \cdot y \cdot \bar{x} = \alpha \cdot \bar{x} \cdot \left[\sum \bar{w}^T \cdot \bar{x}\right]$$

Hebbian Learning Aligns Input and Weight Vectors…

$$\Delta \vec{w} = \alpha \cdot y \cdot \vec{x} \qquad y = \vec{w}^T \cdot \vec{x}$$

(y: scalar (positive))

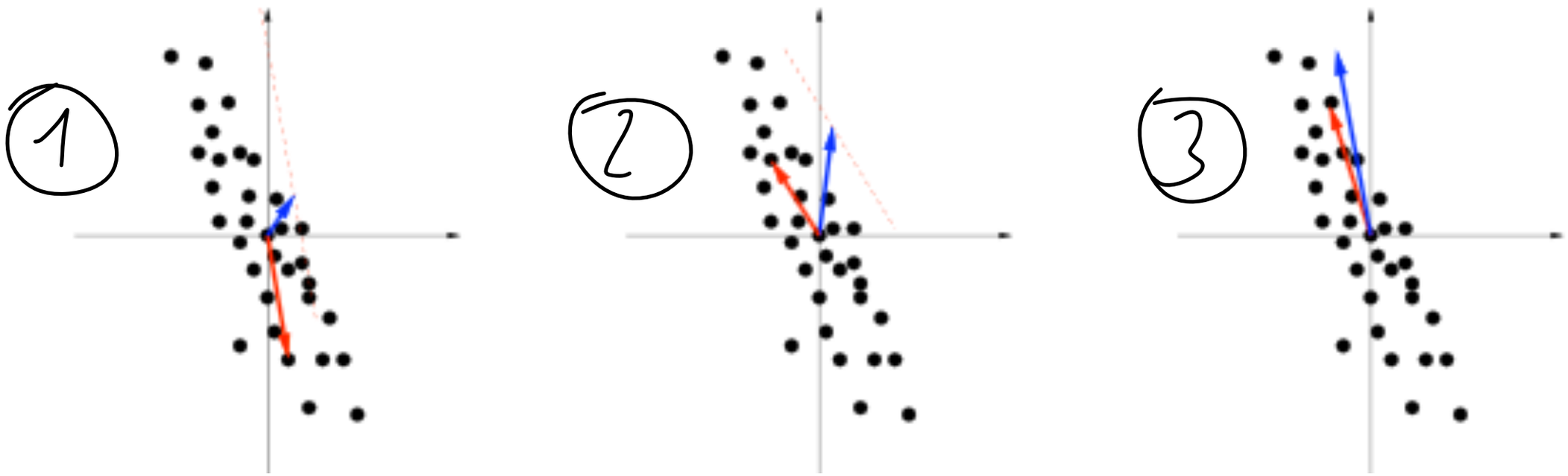
{ 0 if $\vec{w} \perp$ to $\vec{x}$
max. if $\vec{w}$ and $\vec{x}$ are aligned.



Problem: If w points away from the cloud, the dot-product will be negative and both vectors will at the end point towards the opposite direction.

… towards the Direction of Maximum Variability.

HL on Zero-Mean data.



In this case w will ultimately align itself in the direction of the greatest variance!

Hebbian Formalism Expressed as Outer Product.

Form of the Hebbian update rule:

$$\Delta \bar{w} = \alpha\, y \cdot \bar{x}$$

$$= \alpha \left(\bar{w} \cdot \bar{x}\right) \bar{x} \qquad \text{(inner product)}$$

$$= \alpha \left(\bar{x} \cdot \bar{x}^T\right) \cdot \bar{w} \qquad \text{(outer product)}$$

Note on Outer vs. Inner Product.

$$(\bar{v}^T \bar{x})\,\bar{x} = (\bar{x}\,\bar{x}^T)\,\bar{v}$$

$$(\bar{v}^T \bar{x})\cdot\bar{x} = (v_1 x_1 + \dots + v_m x_m)\begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = \begin{bmatrix} v_1 x_1^2 + v_2 x x_2 + \dots + v_m x_1 x_m \\ \vdots \\ v_1 x_m x_1 + v_2 x_m x_1 + \dots + v_m x_m^2 \end{bmatrix}$$

$$= \begin{bmatrix} x_1^2 & x_1 x_2 & \dots & x_1 x_m \\ & & & \\ x_m x_1 & x_m x_2 & \dots & x_m^2 \end{bmatrix}\begin{bmatrix} v_1 \\ \vdots \\ v_m \end{bmatrix} = \boxed{(\bar{x}\cdot\bar{x}^T)\,\bar{v}^T}$$

Hebbian Learning and the Correlation Matrix.

The correlation matrix.

$$\langle \Delta \bar{w} \rangle = \alpha \cdot \langle \bar{x} \bar{x}^T \rangle \bar{w} = \alpha \cdot C \cdot \bar{w}$$

$$C = \begin{bmatrix} x_1^2 & x_1 x_2 \ldots & x_1 x_n \\ & & \\ x_n x_1 & x_2 x_2 \ldots & x_n^2 \end{bmatrix}$$

n is the number of inputs to the Hebbian neuron.

The Problem: Hebbian Learning Blows Up.

The Problem: Hebbian Learning Blows Up.

$$\Delta w = \lambda \cdot C \cdot \bar{v}$$

$$\frac{\partial w}{\partial t} = C\,\bar{w} \longrightarrow$$

Standard Solution $\bar{v} = \bar{u} \cdot e^{\lambda t}$ (λ pos. ??)

$$\frac{\partial w}{\partial t}\left(\bar{u} \cdot e^{\lambda t}\right) = C\left(\bar{u} \cdot e^{\lambda t}\right)$$

$$\lambda \cdot \bar{u} \cdot e^{\lambda t} = C\,\bar{u}\,e^{\lambda t}$$

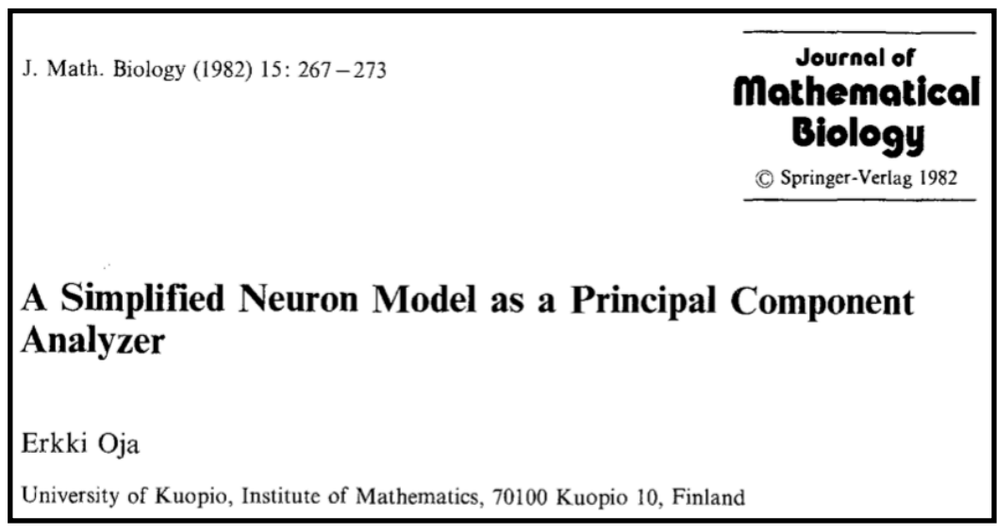
$\Rightarrow$ w blows up

Solution #1: Oja's Learning Rule.

Oja's Solution: Weight Decay.

$$\Delta \vec{w} = \alpha \cdot y \cdot (\vec{x} - y \cdot \vec{w})$$

Important: The weight decay term is proportional to y^2. That means w approaches unit length automatically while still pointing at the direction of maximum variance.



J. Math. Biology (1982) 15: 267–273

Journal of Mathematical Biology

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A Simplified Neuron Model as a Principal Component Analyzer

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Solution #2: The Hebbian Covariance Rule.

Suggested by T. Sanowski:

$$\Delta \bar{\omega} = \alpha \left(y - \langle y \rangle\right) \cdot \left(\bar{x} - \langle \bar{x} \rangle\right)$$

$$\langle \Delta \bar{\omega} \rangle = \alpha \left\langle y\bar{x} - y\langle \bar{x} \rangle - \bar{x}\langle y \rangle + \langle y \rangle \cdot \langle \bar{x} \rangle \right\rangle$$

$$= \alpha \left(\langle y\bar{x} \rangle - \langle y \rangle \langle \bar{x} \rangle - \langle y \rangle \langle \bar{x} \rangle + \langle y \rangle \langle \bar{x} \rangle \right)$$

$$= \alpha \left(\langle y\bar{x} \rangle - \langle y \rangle \langle \bar{x} \rangle \right)$$

$$= \alpha \left(\langle \bar{x}\bar{x}^T \rangle - \langle \bar{x} \rangle \langle \bar{x}^T \rangle \right) \bar{\omega}$$

Hebbian Learning Rules - Conclusions.

We've shown that for zero-mean data, the weight vector aligns itself with the direction of maximum variance as training continues.

This direction corresponds to the eigenvector of the correlation matrix $\mathbf{C}$ with the largest corresponging eigenvalue. If we decompose the current $\bar{w}$ in terms of the eigenvectors of $\mathbf{C}$, $\bar{w} = \sum_i \alpha_i \, \bar{u}_i$, then the expected weight update rule

$$\begin{aligned} \langle \Delta \bar{w} \rangle &= \eta \, \mathbf{C} \, \bar{w} \\ &= \eta \, \mathbf{C} \sum_i \alpha_i \, \bar{u}_i \\ &= \eta \sum_i \alpha_i \, \lambda_i \, \bar{u}_i \end{aligned} \tag{12}$$

will move the weight vector $\bar{w}$ towards eigenvector $\bar{u}_i$ by a factor proportional to λ_i. Over many updates, the eigenvector with the largest λ_i will drown out the contributions from the others.

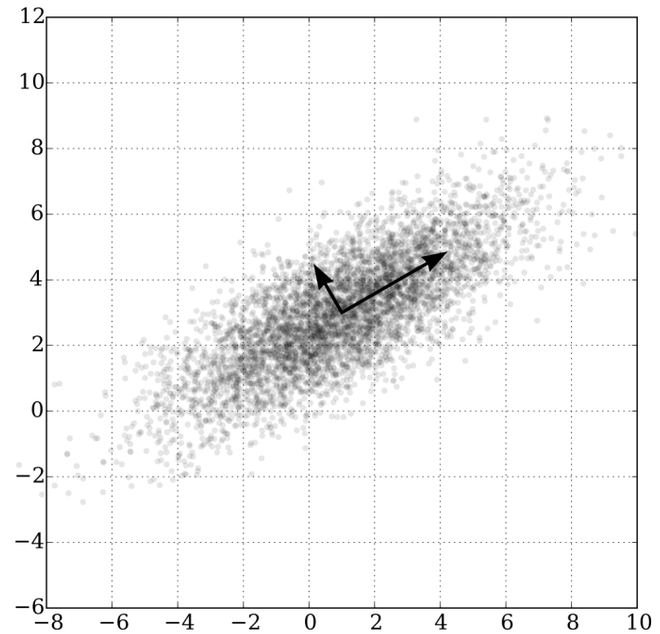
But because of this mechanism, the magnitude of $\bar{w}$ blows up.

A Short Recap of PCA.

A tool from statistics for data analysis.

Can reveal structure in high-N-dimensional data that is not otherwise obvious.

Like Hebbian learning, it discovers the direction of maximum variance in the data. But then in the $(N-1)$-dimensional subspace perpendicular to that direction, it discovers the direction of maximum *remaining* variance, and so on for all N.



The result is an ordered sequence of principal components. These are equivalently the eigenvectors of the correlation matrix $\mathbf{C}$ for zero-mean data, ordered by magnitude of eigenvalue in descending order. They are mutually orthogonal.

Question: Is Hebbian Learning equivalent to PCA? Are there any differences? Can we built a neuronal network that performs PCA?

DISCUSS!

PCA by Neuronal Networks.

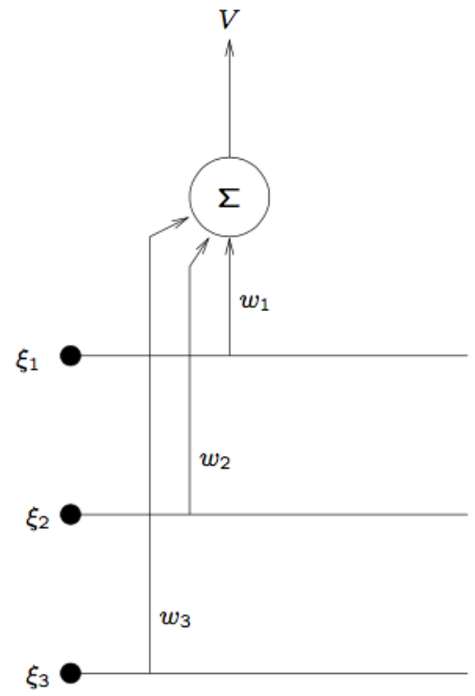
We've seen that Hebbian learning, with appropriate provisions for preventing blow up, extracts the largest principal component.

Let's take a look at two different neural network architectures capable of extracting more of them.

1. Cascading multiple Hebbian neurons.

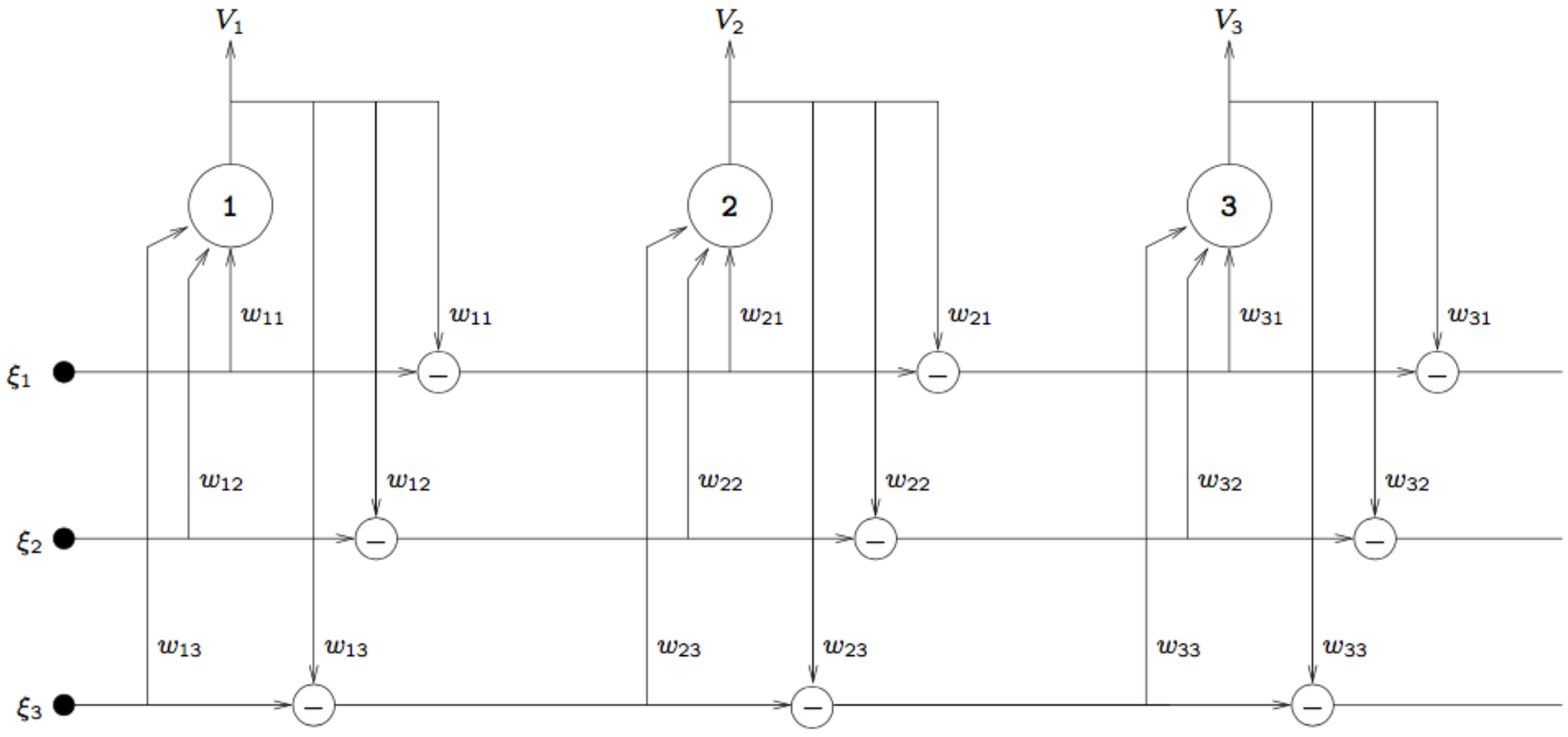
2. Autoencoder networks (we've already seen these before).

An alternate diagram of a Hebbian neuron:



PCA by Neuronal Networks.

PCA Neuronal Network



PCA by Neuronal Networks - Sanger's Rule.

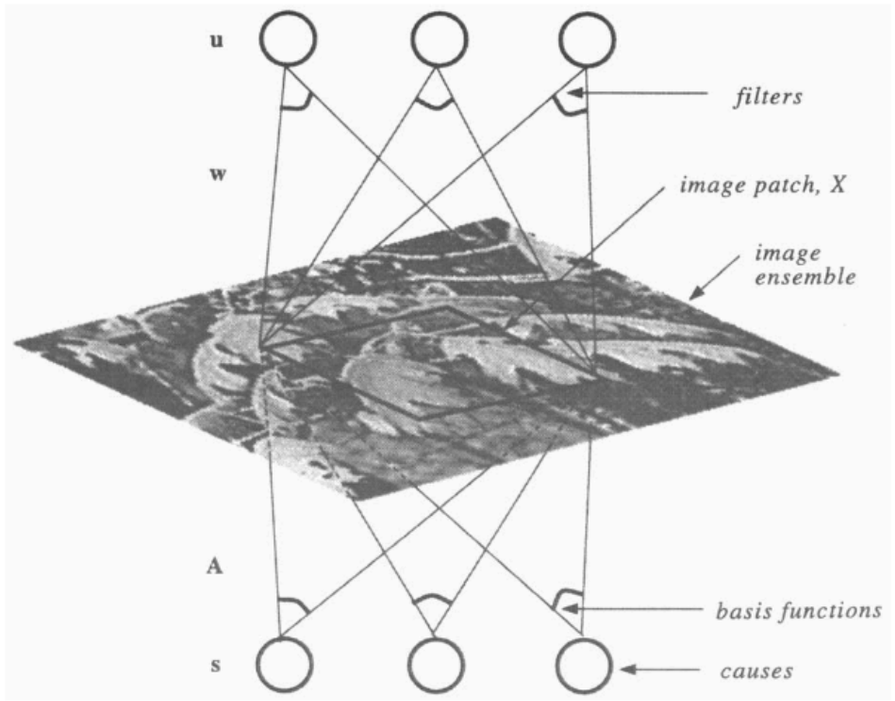
Algorithm:

1. Subtract the contribution of the first principal component.
2. Drive the difference into another Hebbian neuron.
3. This extracts the next principal component.
4. Subtract its contribution. Goto step 2.

With N Hebbian neurons, we'll get all N principal components.

Sejnowski's Infomax Network (ICA)

A. J. BELL and T. J. SEJNOWSKI



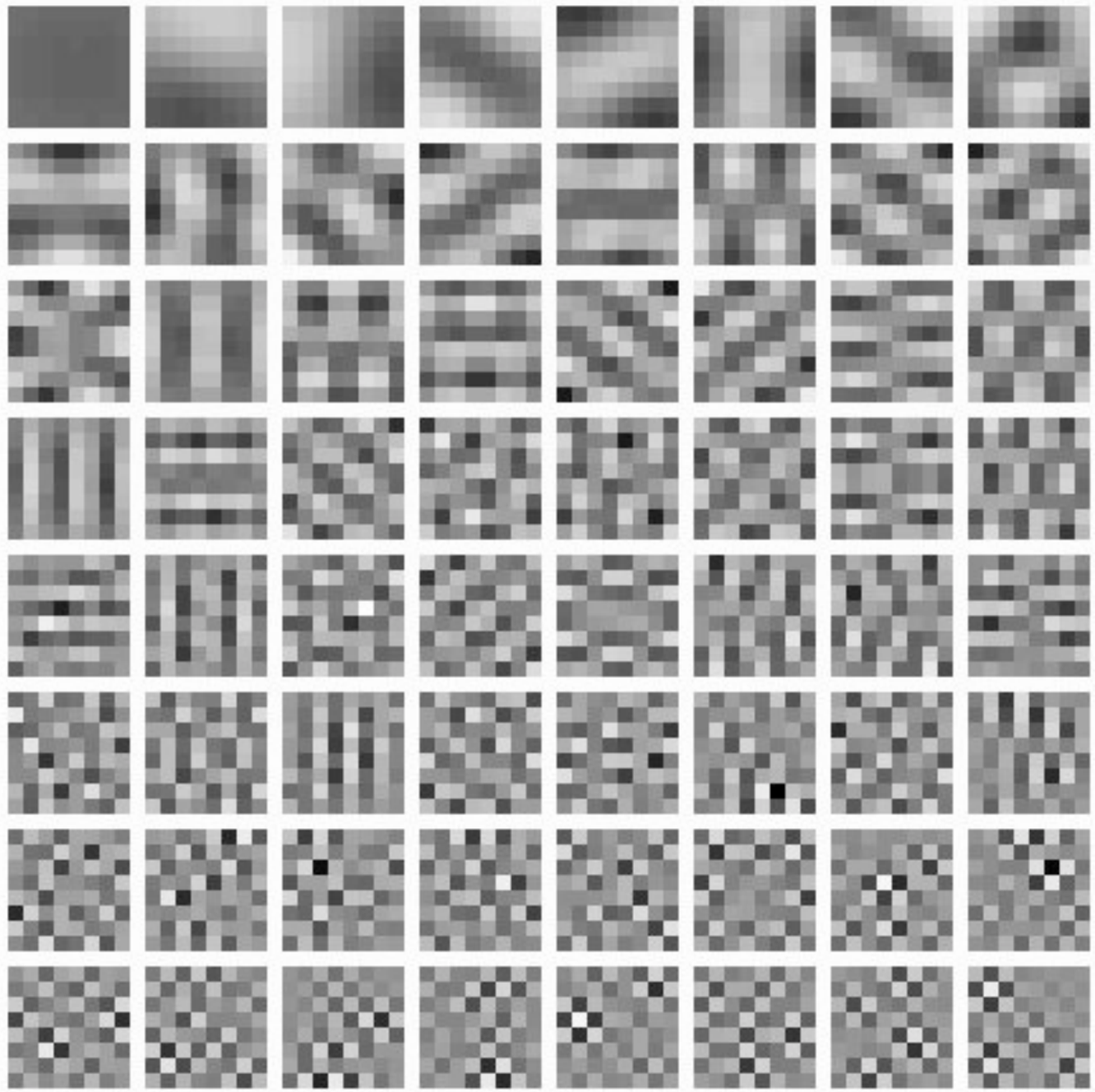
An information-maximisation approach to blind separation and blind deconvolution

Anthony J. Bell and Terrence J. Sejnowski

Computational Neurobiology Laboratory
The Salk Institute
10010 N. Torrey Pines Road
La Jolla, California 92037*

Abstract

We derive a new self-organising learning algorithm which maximises the information transferred in a network of non-linear units. The algorithm does not assume any knowledge of the input distributions, and is defined here for the zero-noise limit. Under these conditions, information maximisation has extra properties not found in the linear case (Linsker 1989). The non-linearities in the transfer function are able to pick up higher-order moments of the input distributions and perform something akin to true redundancy reduction between units in the output representation. This enables the network to separate statistically



PCs from natural images

The BCM (Bienenstock, Cooper, Munro) Rule

THEORY FOR THE DEVELOPMENT OF NEURON SELECTIVITY: ORIENTATION SPECIFICITY AND BINOCULAR INTERACTION IN VISUAL CORTEX[1]

ELIE L. BIENENSTOCK,[2] LEON N COOPER,[3] AND PAUL W. MUNRO

Center for Neural Science, Department of Physics, and Division of Applied Mathematics, Brown University, Providence, Rhode Island 02912

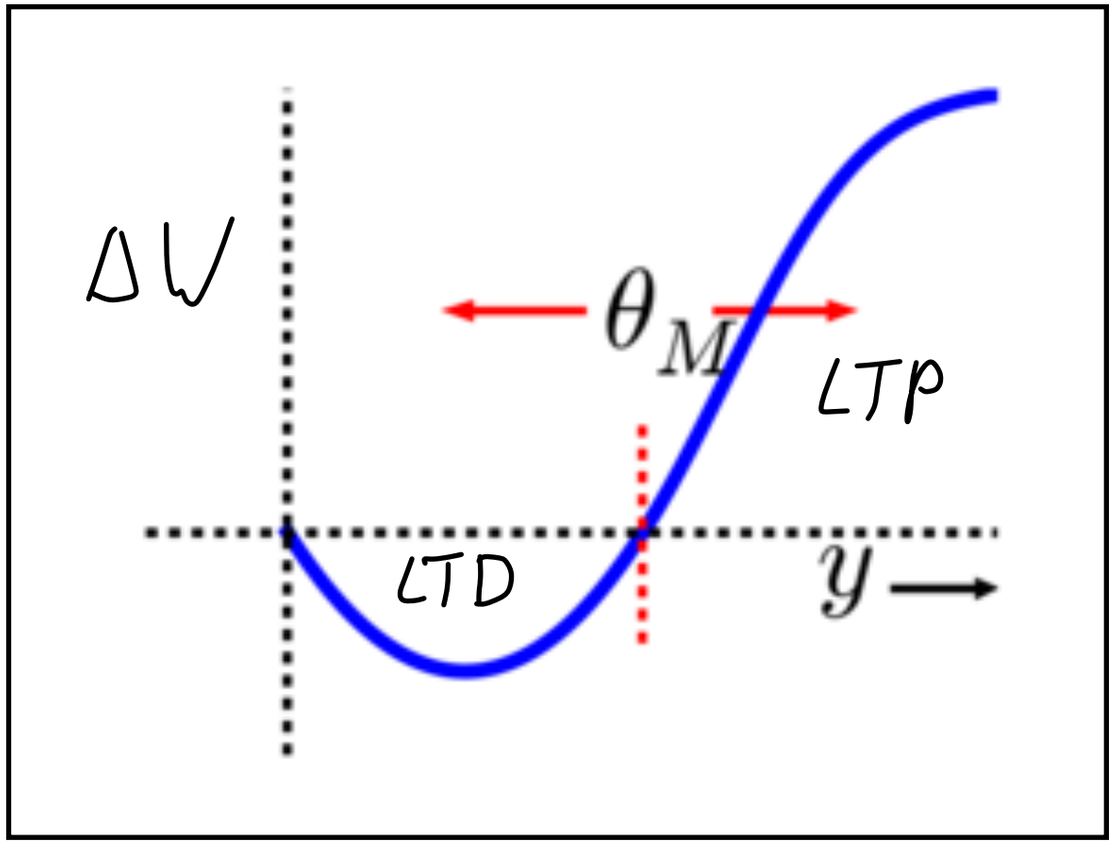
Received June 5, 1981; Revised August 27, 1981; Accepted September 1, 1981

Linear BCM

$$y = \sum_i w_i x_i$$

$$\frac{dw_i}{dt} = y(y - \theta_M)x_i - \epsilon w_i$$

$$\theta_M = E^p[(y/y_o)]$$



The BCM (Bienenstock, Cooper, Munro) Rule

Non-Linear BCM (I-BCM)

$$y = \sigma\left(\sum_i w_i x_i\right)$$

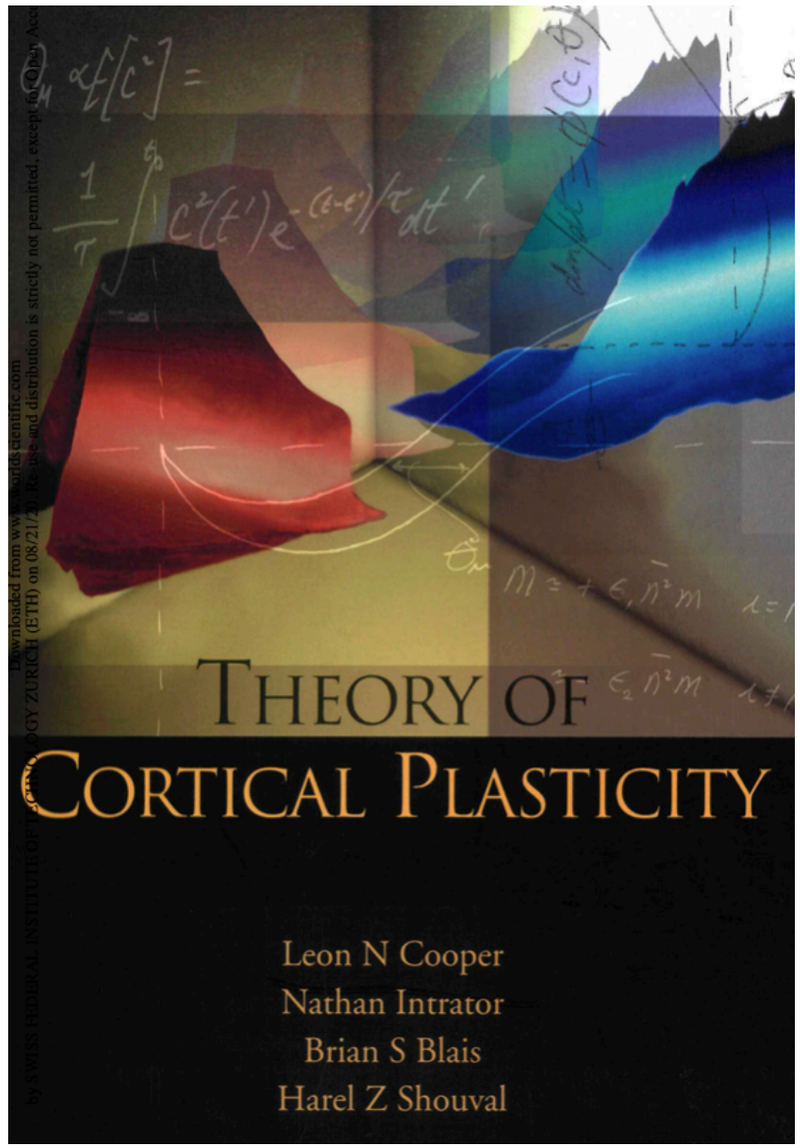
$$\frac{dw_i}{dt} = y(y - \theta_M)x_i\sigma'(y)$$

$$\theta_M = E[y^2]$$

Energy Function:

$$R = -\frac{1}{3}E[y^3] + \frac{1}{4}E^2[y^2]$$

Intrator and Cooper, 1992



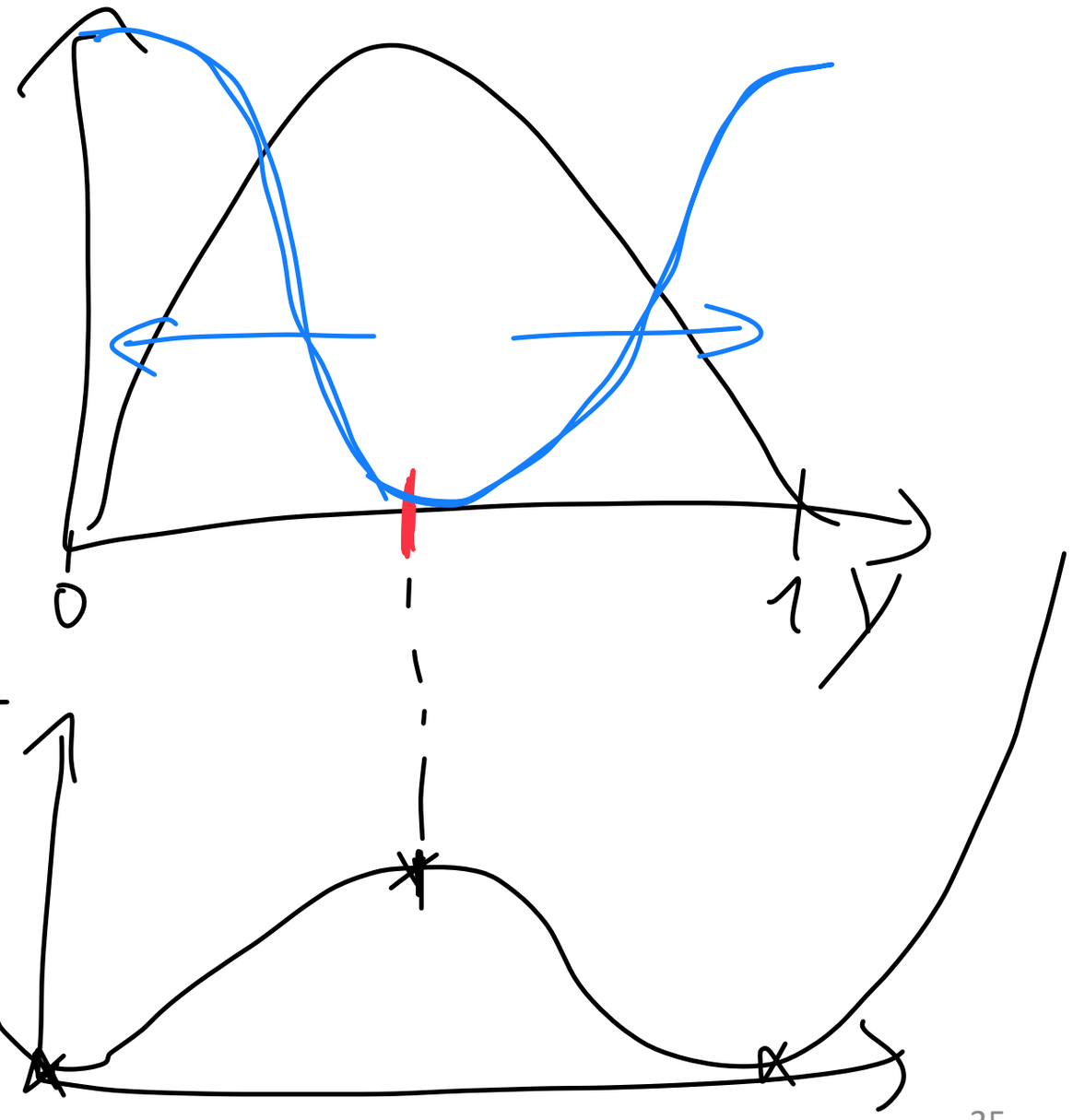
The BCM (Bienenstock, Cooper, Munro) Rule

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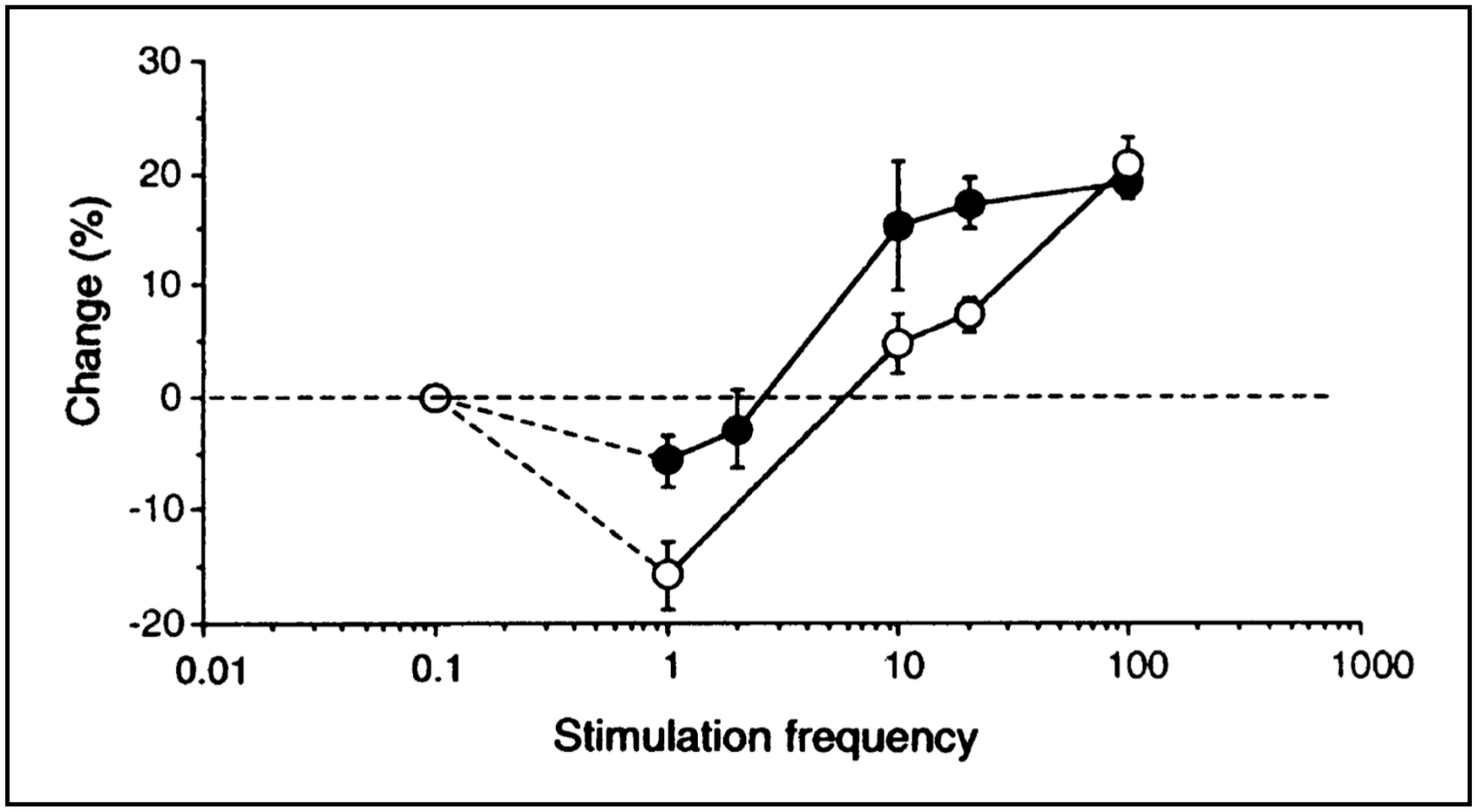
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Experimental Evidence Supporting the BCM Rule



LETTERS TO NATURE

Experience-dependent modification of synaptic plasticity in visual cortex

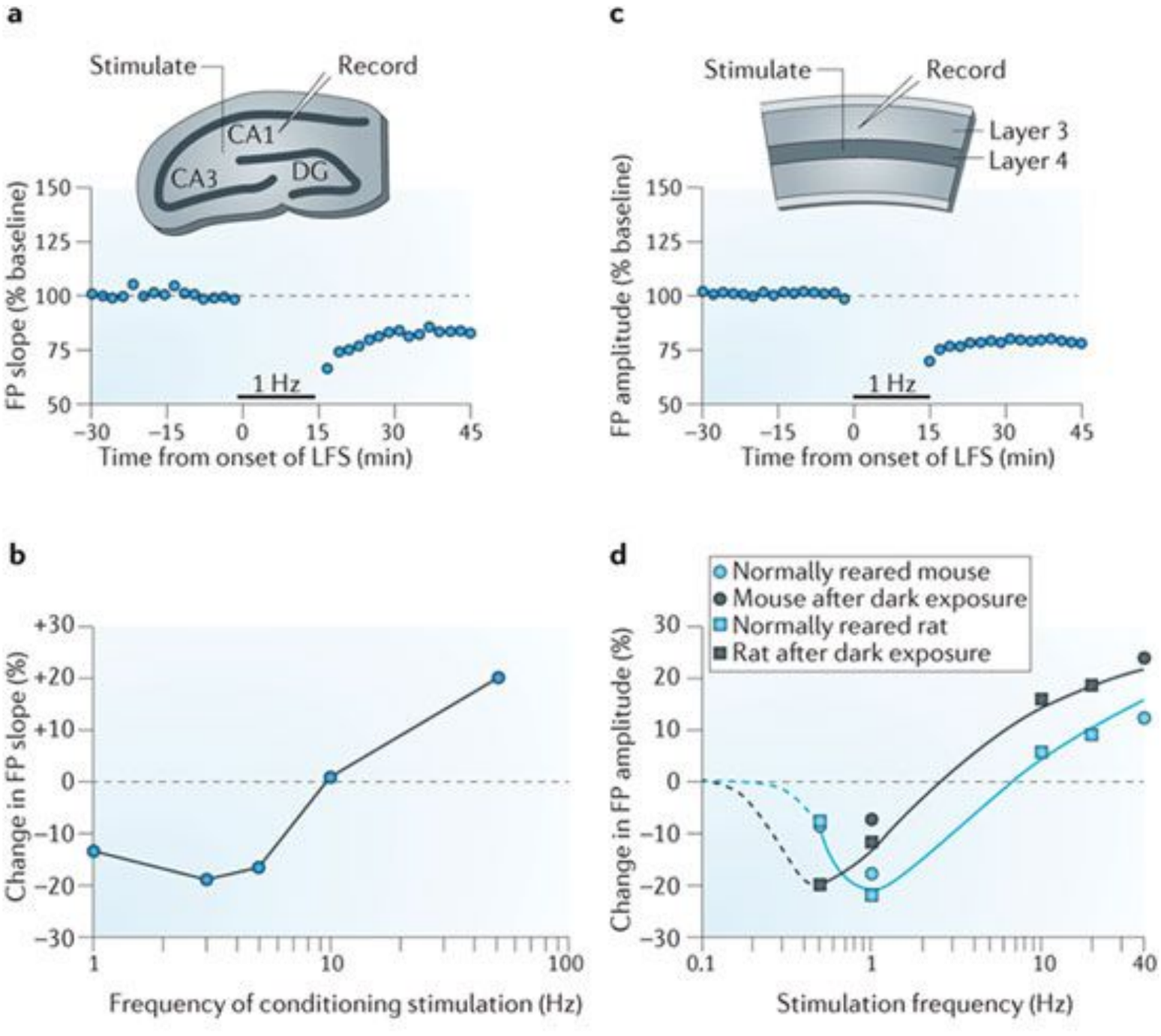
Alfredo Kirkwood, Marc G. Rioult & Mark F. Bear

Department of Neuroscience and Howard Hughes Medical Institute, Brown University, Providence, Rhode Island 02912, USA

Experimental Evidence Supporting the BCM Rule

Hippocampus CA1

Primary Visual Cortex



Opinion | Published: 19 October 2012

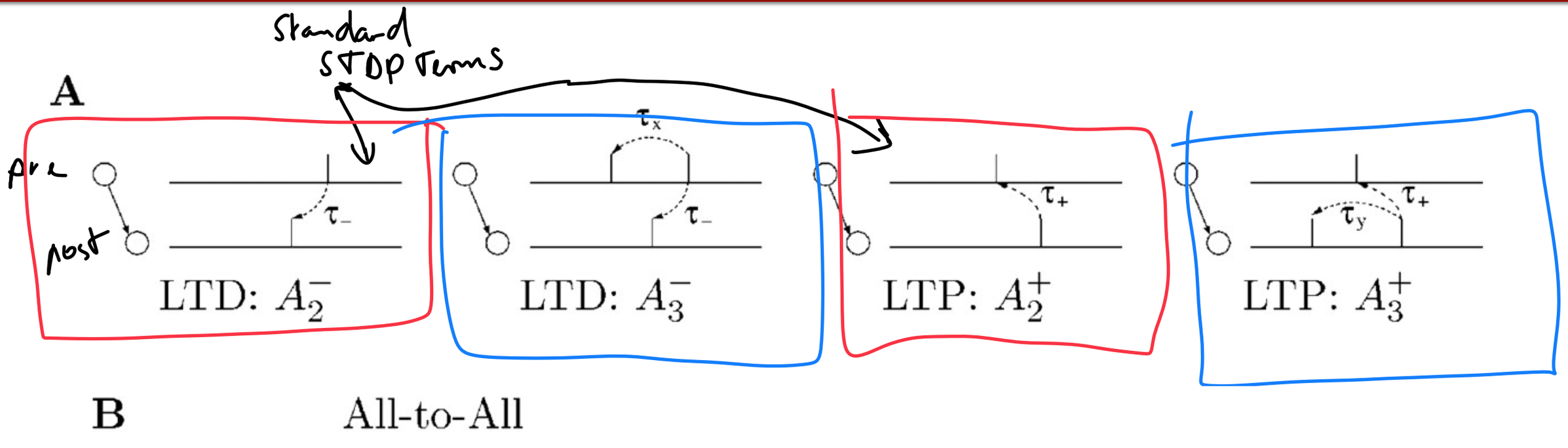
The BCM theory of synapse modification at 30: interaction of theory with experiment

Leon N Cooper & Mark F. Bear

Nature Reviews Neuroscience **13**, 798–810 (2012) | Download Citation

639 Accesses | **136** Citations | **18** Altmetric | Metrics »

The Triplet Rule (STDP for multiple Spikes).

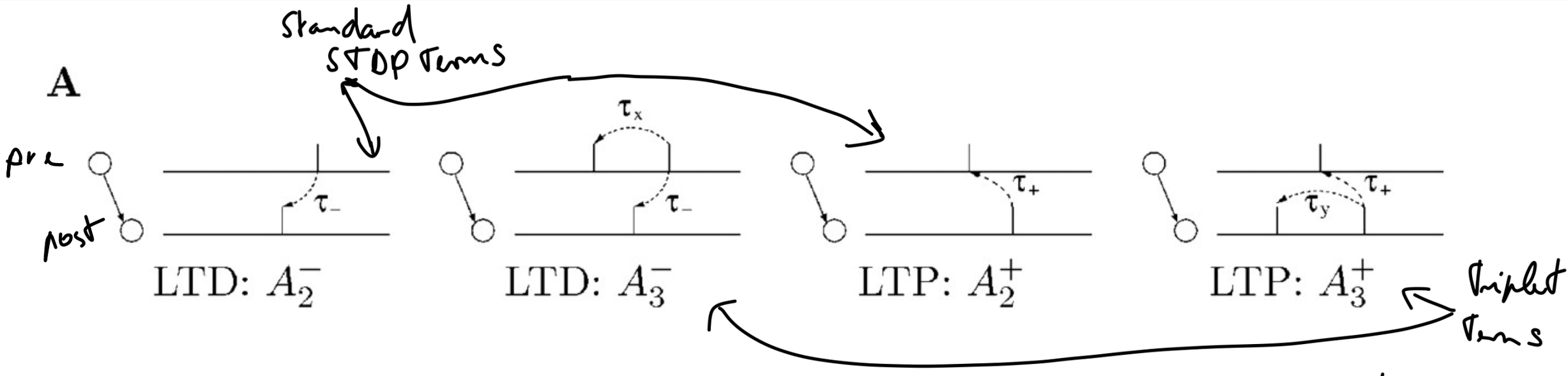


Triplets of Spikes in a Model of Spike Timing-Dependent Plasticity

The Journal of Neuroscience, September 20, 2006 • 26(38):9673–9682 • **9673**

Jean-Pascal Pfister and Wulfram Gerstner
Laboratory of Computational Neuroscience, School of Computer and Communication Sciences and Brain-Mind Institute, Ecole Polytechnique Fédérale de Lausanne, CH-1015 Lausanne, Switzerland

The Triplet Rule.



$$w(t) \rightarrow w(t) - o_1(t)[A_2^- + A_3^- r_2(t-\epsilon)] \text{ if } t = t^{\text{pre}}. \rightarrow \text{pre active}$$

$$w(t) \rightarrow w(t) + r_1(t)[A_2^+ + A_3^+ o_2(t-\epsilon)] \text{ if } t = t^{\text{post}}. \rightarrow \text{post active}$$

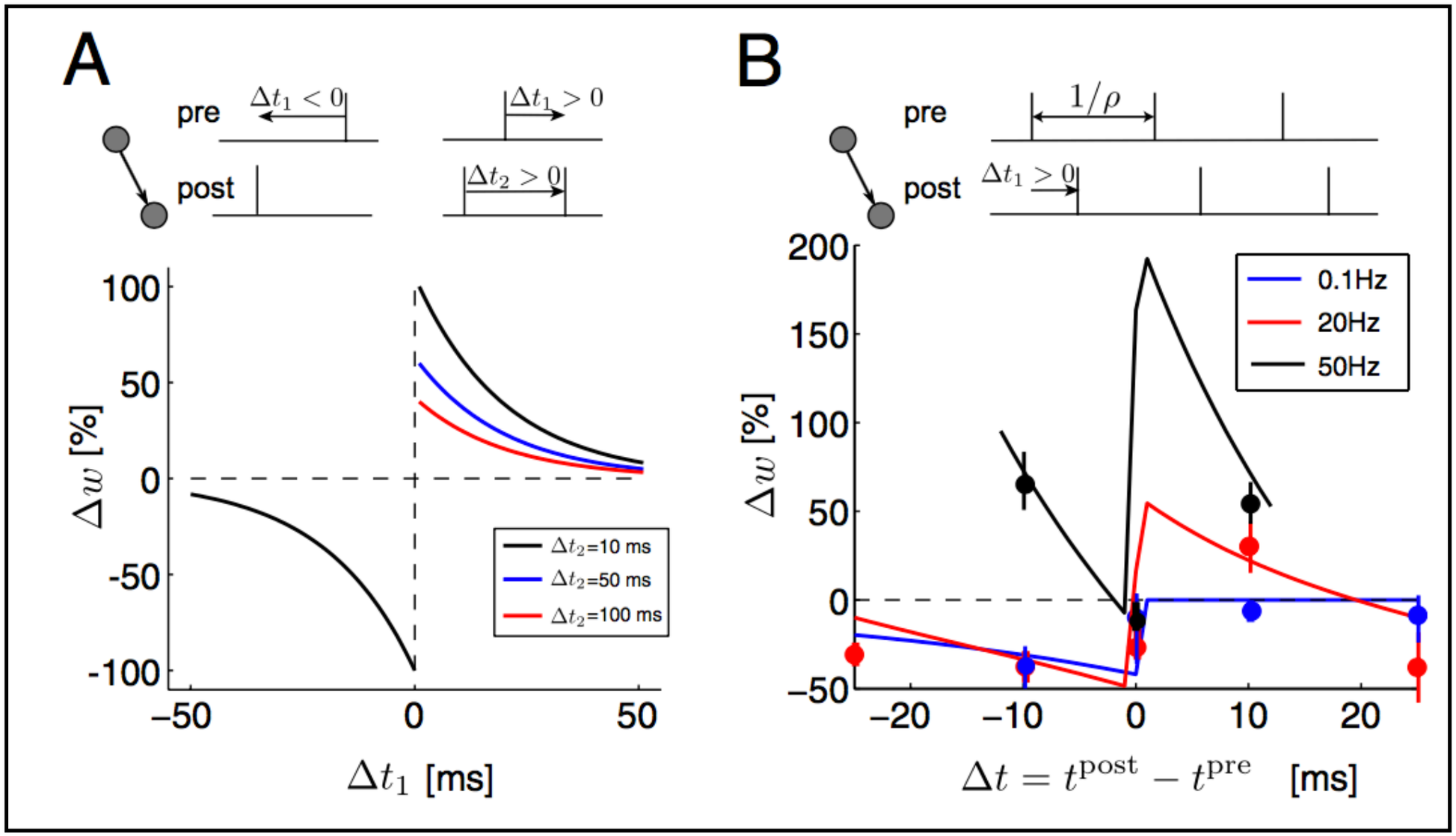
Note that if we set A3+ = 0 and A3− = 0, the model becomes a classical pair-based STDP model

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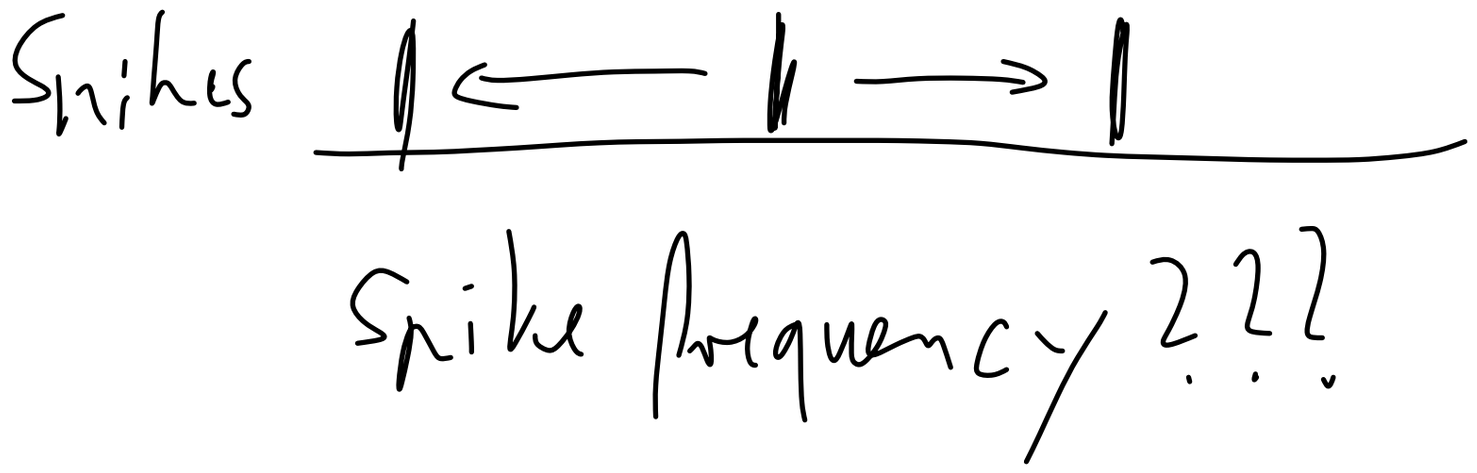
Can the Triplet Rule be Mapped onto the BCM Rule?



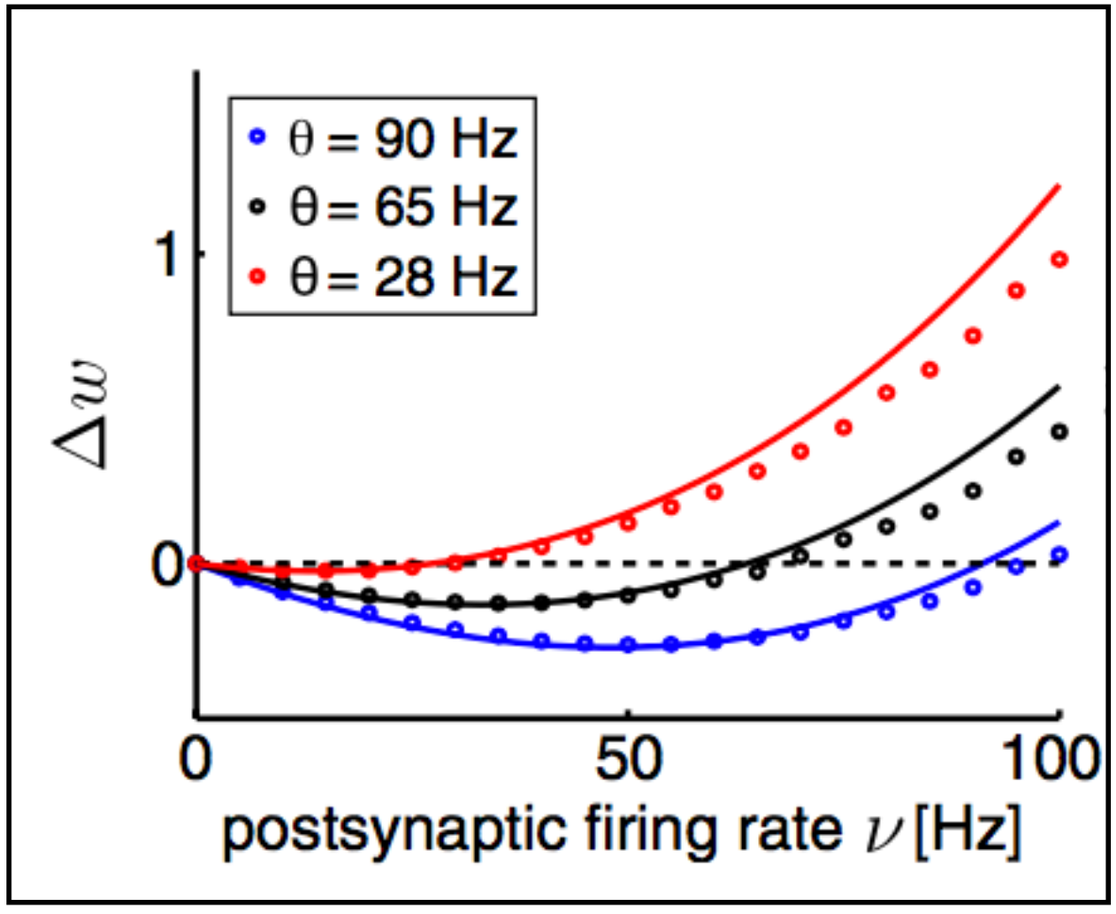
Can the Triplet Rule be Mapped onto the BCM Rule?

How can we relate the BCM Rule to the Triplet Rule?

DISCUSS!



Can the Triplet Rule be Mapped to the BCM Rule?



A triplet spike-timing–dependent plasticity model generalizes the Bienenstock–Cooper–Munro rule to higher-order spatiotemporal correlations

Julijana Gjorgjieva[a,1,2], Claudia Clopath[b], Juliette Audet[c], and Jean-Pascal Pfister[d,e]

www.pnas.org/cgi/doi/10.1073/pnas.1105933108

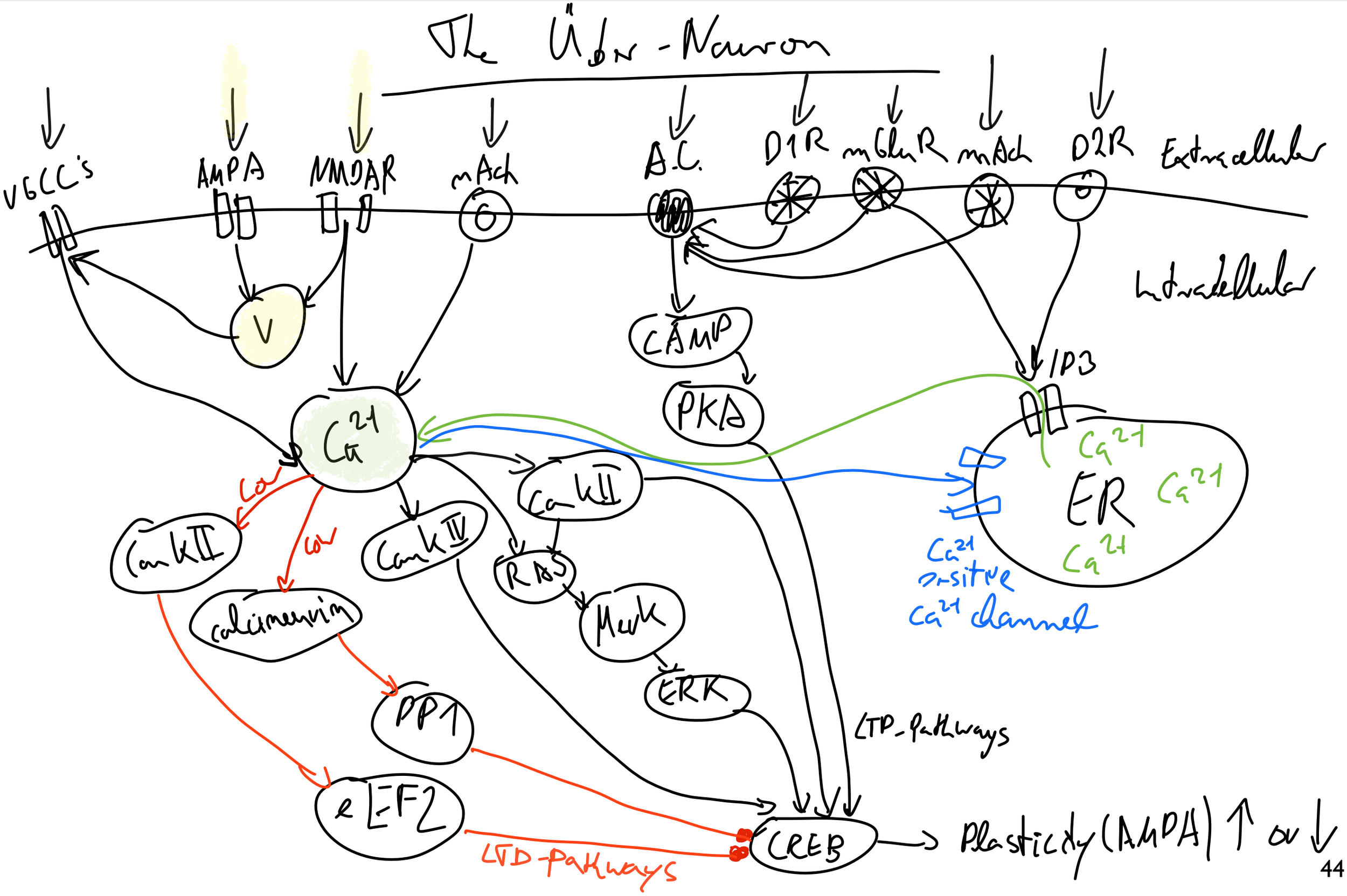
The Calcium Rule.

Calcium-based plasticity model explains sensitivity of synaptic changes to spike pattern, rate, and dendritic location

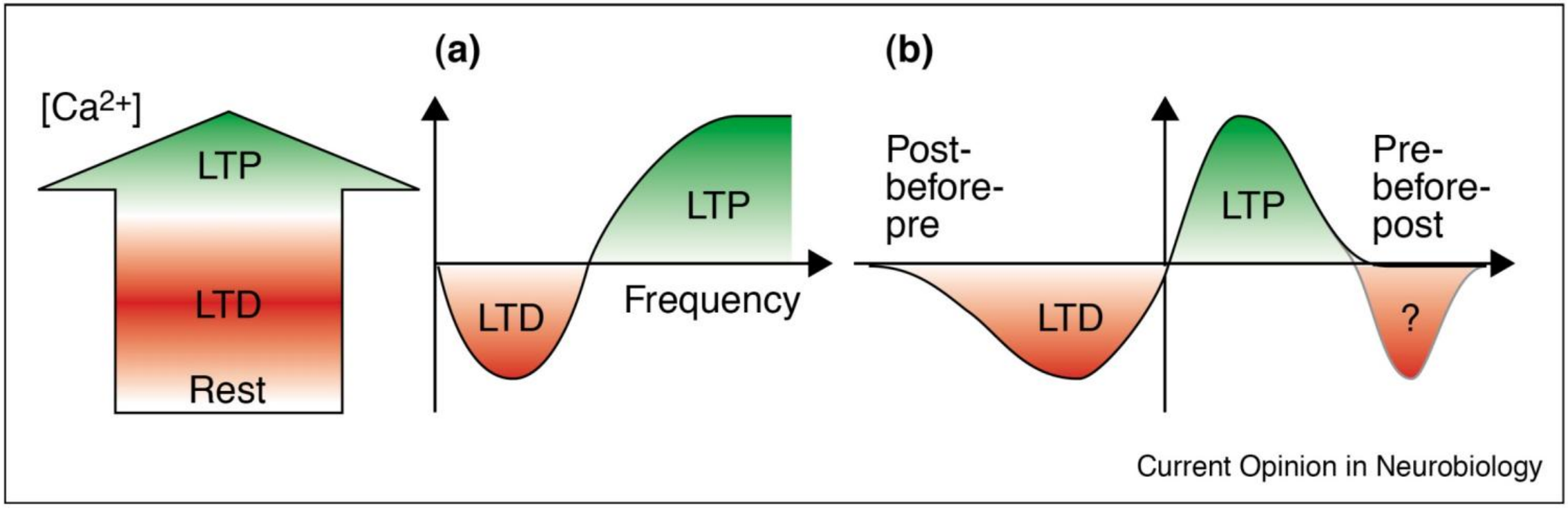
Michael Graupner[a,b,1] and Nicolas Brunel[a]

www.pnas.org/cgi/doi/10.1073/pnas.1109359109

Intracellular Plasticity Signaling Pathways.



Synaptic Plasticity Strongly Depends on Calcium Levels.

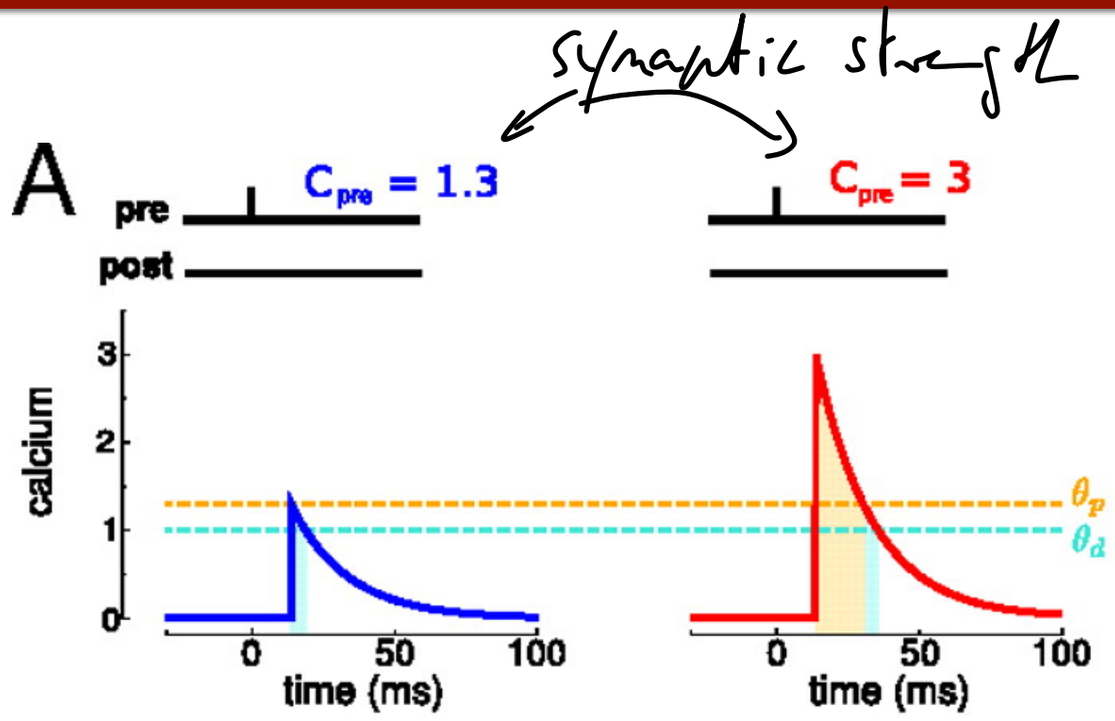


1. Level and timing of Ca^{2+} rise in spine determines LTD or LTP.

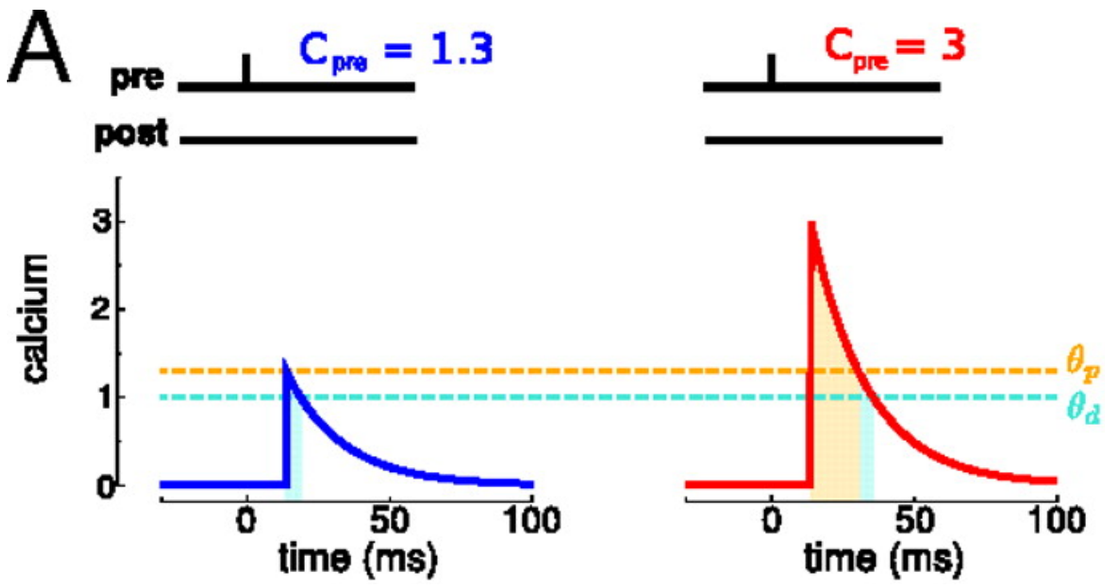
2. Low frequency synaptic firing (~5 Hz) produces LTD;
high frequency synaptic firing (~50 to 100 Hz) produces LTP.

3. The same Ca^{2+} rules may underlie
“spike-timing-dependent synaptic plasticity (STDP).

The Calcium Rule.



The Calcium Rule.



$$\tau\frac{d\rho}{dt} = -\rho(1-\rho)(\rho_\star-\rho) + \gamma_p(1-\rho)\Theta[c(t)-\theta_p] - \gamma_d\rho\Theta[c(t)-\theta_d] + Noise(t).$$

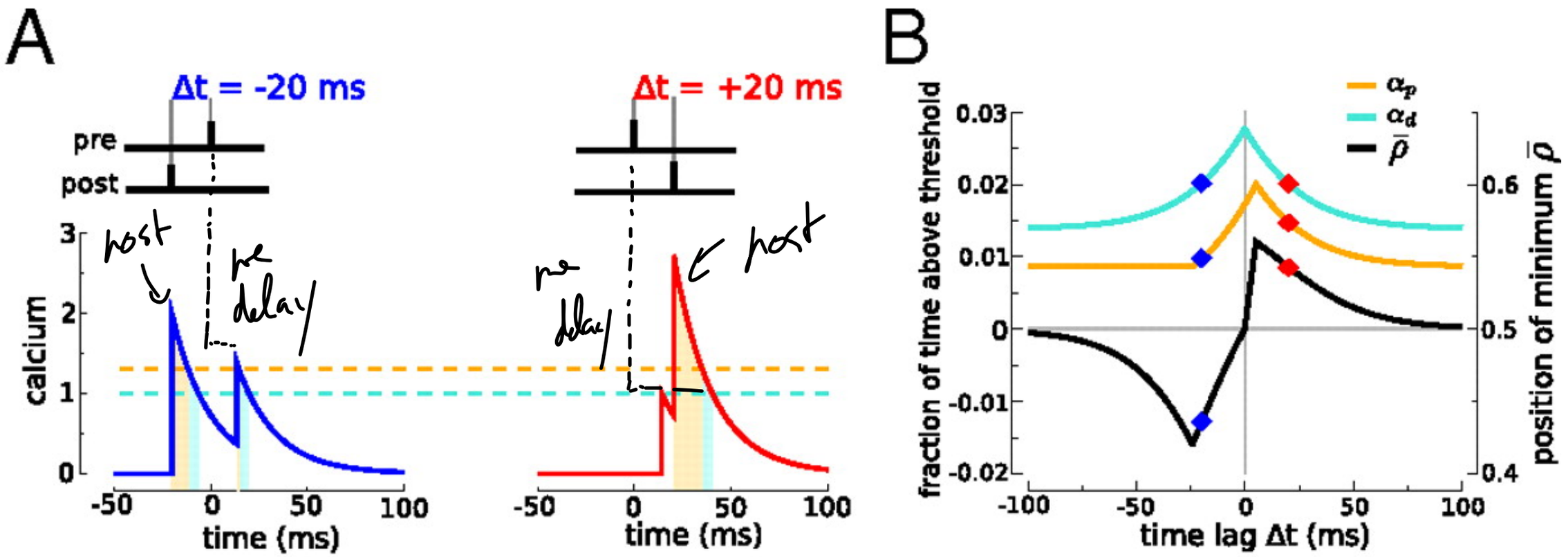
for no act. ρ goes to 0 or 1

Calcium dependency

Heaviside step function.

γ_p, γ_d: learning rates

The Calcium Rule.



The Problem with Hebbian Learning!

The Ups and Downs of Hebb Synapses

GEOFFREY HINTON
University of Toronto

Abstract

Modelers have come up with many different learning rules for neural networks. When a teacher specifies the correct output, error-driven rules work better than pure Hebb rules in which the changes in synapse strength depend on the correlation between pre and postsynaptic activities. But for unsupervised learning, Hebb rules can be very effective if they are combined with suitable normalization or "unlearning" terms to prevent the synapses growing without bound. Hebb rules that use rates of change of activity instead of activity itself are useful for discovering perceptual invariants and may also provide a way of implementing error-driven learning.

This approach led to effective "error-driven" learning rules such as the Widrow-Hoff rule (Widrow & Hoff, 1960) and the perceptron convergence procedure (Rosenblatt, 1961) and it was later generalized to multilayer networks by using backpropagation of the errors to get training signals for intermediate "hidden" layers (Rumelhart, Hinton, & Williams, 1986). Within the neural network community, the "Hebbian" approach of using the product of pre and postsynaptic activities to drive learning was seen as inferior to error-driven methods that use the product of the presynaptic activity and the postsynaptic activity *derivative* – the rate at which the objective function changes as the postsynaptic activity is changed. Even when the task was merely to associate random input vectors with random output vectors, it was shown that an error-driven rule worked much better than a Hebbian rule.

Unfortunately, error-driven learning has come seri-

What was Geoffrey's Problem with Hebbian Learning?

$$\dot{w} = H(\text{pre}, \text{post})$$

Hinton 2003,
http://dx.doi.org/10.1037/

Hebbian Learning Implements Unsupervised Learning!

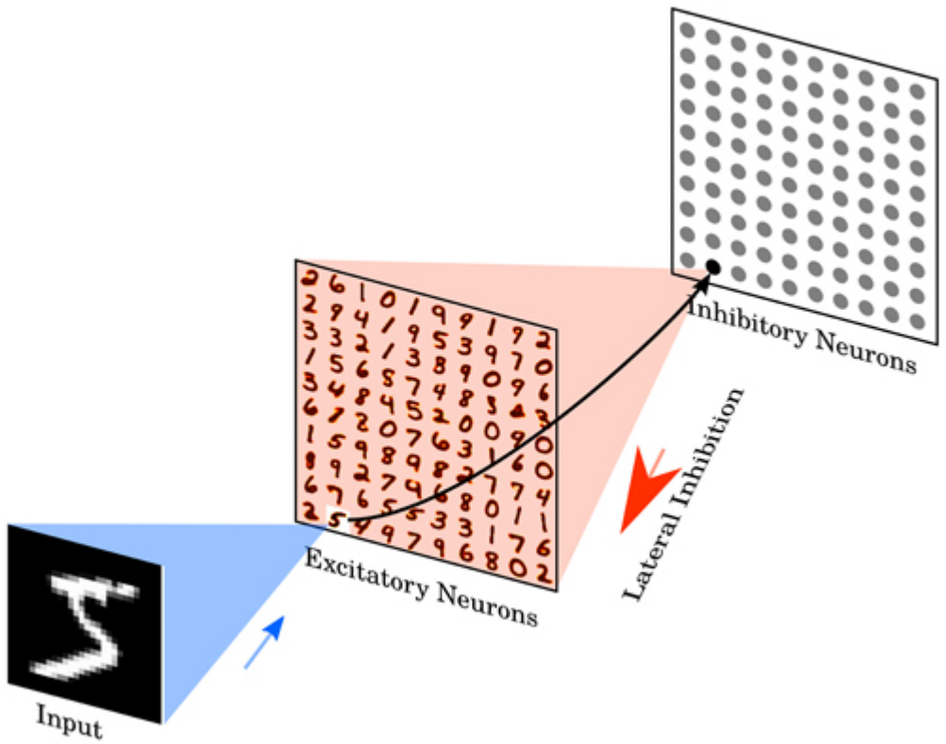
ORIGINAL RESEARCH ARTICLE

Front. Comput. Neurosci., 03 August 2015 | https://doi.org/10.3389/fncom.2015.00099

Unsupervised learning of digit recognition using spike-timing-dependent plasticity

 Peter U. Diehl* and **Matthew Cook**

Institute of Neuroinformatics, ETH Zurich and University Zurich, Zurich, Switzerland



(1) Use UL to learn digit clusters

(2) Use supervised learning on top of the clustering

(3) The better your clustering the faster/better the test performance.

The Solution: Three Factor Hebbian Learning Rules!

Front Neural Circuits. 2015; 9: 85.
Published online 2016 Jan 19. doi: 10.3389/fncir.2015.00085
PMCID: PMC4717313
PMID: 26834568

Neuromodulated Spike-Timing-Dependent Plasticity, and Theory of Three-Factor Learning Rules

Nicolas Frémaux and Wulfram Gerstner*

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$$\dot{w} = H\,(\text{pre}, \text{post})$$

$$\dot{w} = F\,(M, \text{pre}, \text{post})$$

The Solution: Three Factor Hebbian Learning Rules!

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$$\dot{w} = F(M, \text{pre}, \text{post})$$

What could M be in a biological Neuronal Network?

Discuss!

The Solution: Three Factor Hebbian Learning Rules!

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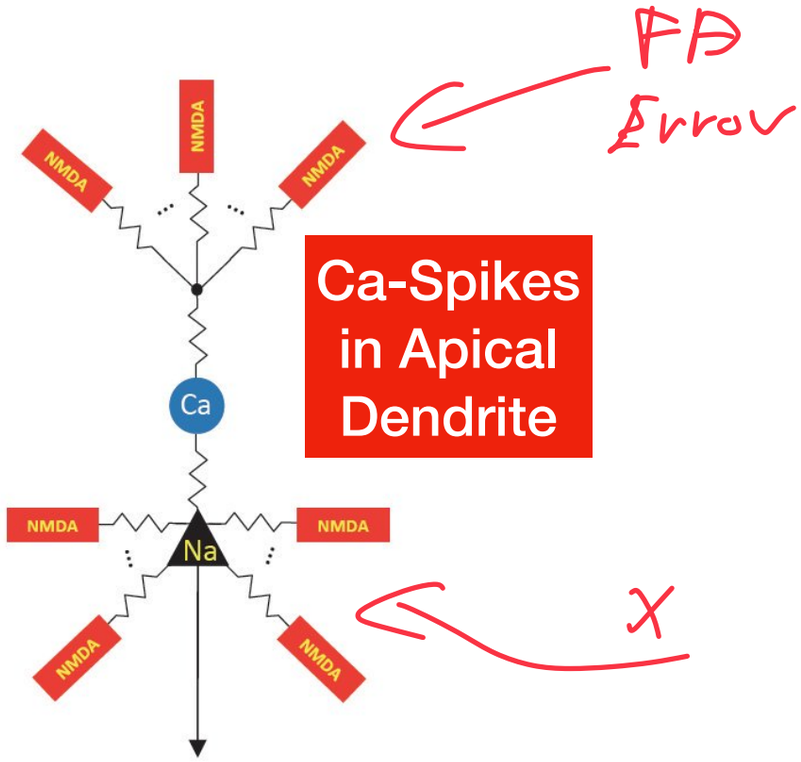
Extracellular calcium release from astrocytes

Neuromodulators NE/DA/ACh

Predictive signal in Time

Activity modulation through inhibition

M could be of the form $(t_i - y_i)$



Lecture Recap. What did we learn today?

1. Learning Rules:
   - 1.1) Why (local) neuronal learning rules important?
   - 1.2) The Perceptron Learning Rule
   - 1.3) ADALINE and the Delta rule
   - 1.4) Recap of Hebbian learning
   - 1.5) Oja's Rule
   - 1.6) The Covariance Rule
   - 1.7) Sanger's Rule
   - 1.8) The Calcium Rule
   - 1.9) The Bienenstock-Cooper-Monroe (BCM) rule.
   - 1.10) The Triplet Rule (Pfister&Gerstner).
2. Extensions of hebbian learning rules (neo Hebbian)
   - 2.1. What is a bio-plausible 3rd factor (M)?

Thank you for your Attention!

